CUMath: A Benchmark for Evaluating LLM Step-by-Step Reasoning in Undergraduate Mathematics

Quyen Tran[1], Hieu Tran[2], Sutthirut Charoenphon[1], Brian T. Howard[1]

[1]DePauw University, [2]Purdue University

Abstract

Mathematical reasoning has been playing a crucial role in assessing the capabilities of Large Language Models (LLMs). Recent findings show that LLMs perform well in deriving the correct answers from inaccurate reasoning steps. This suggests that solution annotations in existing datasets are limited, while hinting at the overly simple or excessively difficult nature of these benchmarks. To address this, we introduce CUMath, a benchmark of 2,100 real problems from undergraduate courses with step-by-step solutions to enable evaluation of both final answers and intermediate reasoning. Moreover, current evaluations treat accuracy and reasoning separately, overlooking their joint role in problem-solving. We therefore evaluate 15 LLMs across various prompting strategies under a step-aware evaluation framework. Our results show that even advanced models often misuse symbolic methods and rely on shortcuts, leading to polished but flawed solutions. These findings highlight persistent inconsistencies in mathematical reasoning and underscore the need for improved benchmarks, evaluation frameworks, and models with stronger reasoning consistency.

1 Introduction

With the rapid development of Large Language Models (LLMs) in solving high-complex mathematical problems, investigating LLMs' mathematical reasoning has been increasingly important (Ahn et al., 2024; Liu et al., 2025). Various techniques (e.g., prompting (Wei et al., 2022; Yao et al., 2023; Wang et al., 2023a), reinforcement learning (Wang et al., 2025a,c; Hubert et al., 2025), test-time scaling (Wu et al., 2024; Snell et al., 2024)) have been explored to enchance the reasoning capabilities. The results showed that LLMs have the abilities to perform sequential deduction to reach the correct final answer (Wang et al., 2025b). However, despite the correct answers, these step-by-step solutions could be erroneous and hallucinating, which hinder the users with limited mathematical knowledge of the refined writings (Boye and Moell, 2025).

This shortcoming potentially comes from the lack of annotation in step-by-step reasoning. The majority of recent benchmarks (e.g., MATH (Hendrycks et al., 2021b) and UGMath (Xu et al., 2025)) focus heavily on evaluating the final answers rather than assessing and quantifying the intermediate steps. Similarly, GSM8k (Cobbe et al., 2021) only represented high-level of free-form intermediate steps-by-steps rather than detailed annotation on the inherent logic sequence of mathematics. In addition, evaluating complex free-form responses poses a challenge to LLMs (Hendrycks et al., 2021a), resulting in LLMs' inconsistency in constructing correct logical order in deriving the correct answers (Zhou et al., 2023).

To address these issues, we introduce **CUMath** benchmarks:

1. **A new benchmark for undergraduate-level mathematical reasoning with step-by-step annotation.** We introduce CUMath, a dataset of 2,100 problems evenly distributed across seven core subjects. A key feature of CUMath is that each problem consistently includes expert-manual step-by-step solutions to quantitatively and qualitatively evaluate the mathematical reasoning capabilities of LLMs.
2. **Multi-metric Evaluation on Step-by-step Mathematical Reasonings.** We combine multiple automatic metrics to evaluate LLM performance on mathematical reasoning, including Exact Match, F1, Stepwise Reasoning Score, and Validity–Redundancy Score. This combined evaluation provides a more comprehensive assessment of both final-answer correctness and the quality of intermediate reasoning steps.
3. **An empirical analysis of LLM reasoning**

Dataset	Source	Levels	%CU	#Types	#Subj.	#Test	%FR	Steps
GSM8k (Cobbe et al., 2021)	Crowdsourced	E	0	1	–	1k	0	No
MATH (Hendrycks et al., 2021b)	Competition	H,O	0	3	7	5k	100	Yes
MiniF2F (Zheng et al., 2022)	Olympiad	E,H,O	0	3	–	244	100	Yes
MathVerse (Zhang et al., 2024)	Public Sources	H	0	3	–	4.7k	45	No
MathVista (Lu et al., 2024)	Mixed Benchmarks	E,H,O	0	3	–	5k	46	No
MathOdyssey (Fang et al., 2024)	Expert-written	H,O,U	$\sim$10	1	–	387	100	No
MMMUMath (Yue et al., 2024)	Mixed Benchmarks	E,H,U	0	1	–	505	0	No
We-Math (Qiao et al., 2024)	Publicly sources	H,U	$\sim$20	3	–	1.7k	100	No
OCWCourses (Lewkowycz et al., 2022)	Public Sources	U	$\sim$18	1	–	272	100	No
UGMathBench (Xu et al., 2025)	Course Website	U	$\sim$50	10	16	5.5k	0	No
CUMath	**Course Materials/ AMS Textbooks**	U	**100**	**3**	**7**	**2.1k**	**$\sim$75**	**Yes**

Table 1: Comparison of math datasets by level (E: Elementary–Middle School, H: High School, O: Olympiad, U: Undergraduate), computational undergraduate coverage (%CU), number of task types, subjects, test size, free-response (FR) proportion, and availability of step-by-step solutions.

gaps. Using CUMath and our framework, we show that LLMs continue to exhibit systematic errors in symbolic manipulation and procedural reasoning, even when producing correct final answers. These findings underscore the importance of evaluating reasoning validity in conjunction with correctness.

2 Related Work

Mathematical Reasoning Datasets. Table 1 shows the variety of datasets in evaluating mathematical reasoning across diverse topics with different difficulty levels. Prior benchmarks were primarily collected from various public sources, raising the possibility of data contamination or hinting at the overlap of data used in the models' training process (Deng et al., 2024). Earlier benchmarks, GSM8K (Cobbe et al., 2021) and MATH (Hendrycks et al., 2021b), focused heavily on elementary or high-school mathematics. MiniF2F (Zheng et al., 2022) targeted at more challenging competition, such as Olympiad-level problems. Multimodal reasoning tasks and diverse problem formats were introduced in recent benchmarks (e.g., MathVista (Lu et al., 2024), MathVerse (Zhang et al., 2024), and MMMU (Yue et al., 2024)). Interestingly, a very limited number of benchmarks address undergraduate-level mathematics, notably, UGMathBench (Xu et al., 2025) and MathOdyssey (Fang et al., 2024). However, these datasets concentrated on elementary problems (e.g., arithmetic and basic algebra) that have been well-covered in the majority of prior datasets (e.g., MATH and GSM8K).

Additionally, most existing datasets focus solely on the final correct answers while overlooking step-by-step reasoning, which offer insufficient evidence to evaluate the quality and correctness of a model's reasoning process. Even though some existing works, such as MATH (Hendrycks et al., 2021b), provide step-by-step solutions, these annotations are often represented at a high level and fail to explicitly capture the logical dependencies between intermediate steps. This creates challenges in assessing the logical consistency and factuality in LLMs' answers, despite the correct answers.

Evaluate Mathematical Reasoning. Understanding the mathematical reasoning of LLMs is essential to further evaluate and improve the capabilities of LLMs (Lee and Hockenmaier, 2025). Precedent works on mathematical reasoning primarily focuses on final-answer accuracy (Hendrycks et al., 2021a; Cobbe et al., 2021); however, the correctness of the final answer does not necessarily imply that the intermediate reasoning is faithful, valid, or logically sound (Li et al., 2024; Farquhar et al., 2024; Shojaee et al., 2025). To better capture the reasoning process, step-by-step reasoning has emerged as an alternative evaluation paradigm, in which LLMs generate a sequence of intermediate natural-language steps before producing the final answer (Wei et al., 2022).

To evaluate intermediate reasoning more directly, prior work has constructed meta-evaluation datasets with annotated reasoning errors (Song et al., 2025) and developed automatic frameworks for scoring reasoning traces at scale (Golovneva et al., 2023; Xia et al., 2025). Common evaluation criteria can be grouped into four main types: factuality, validity, coherence, and utility (Lee and

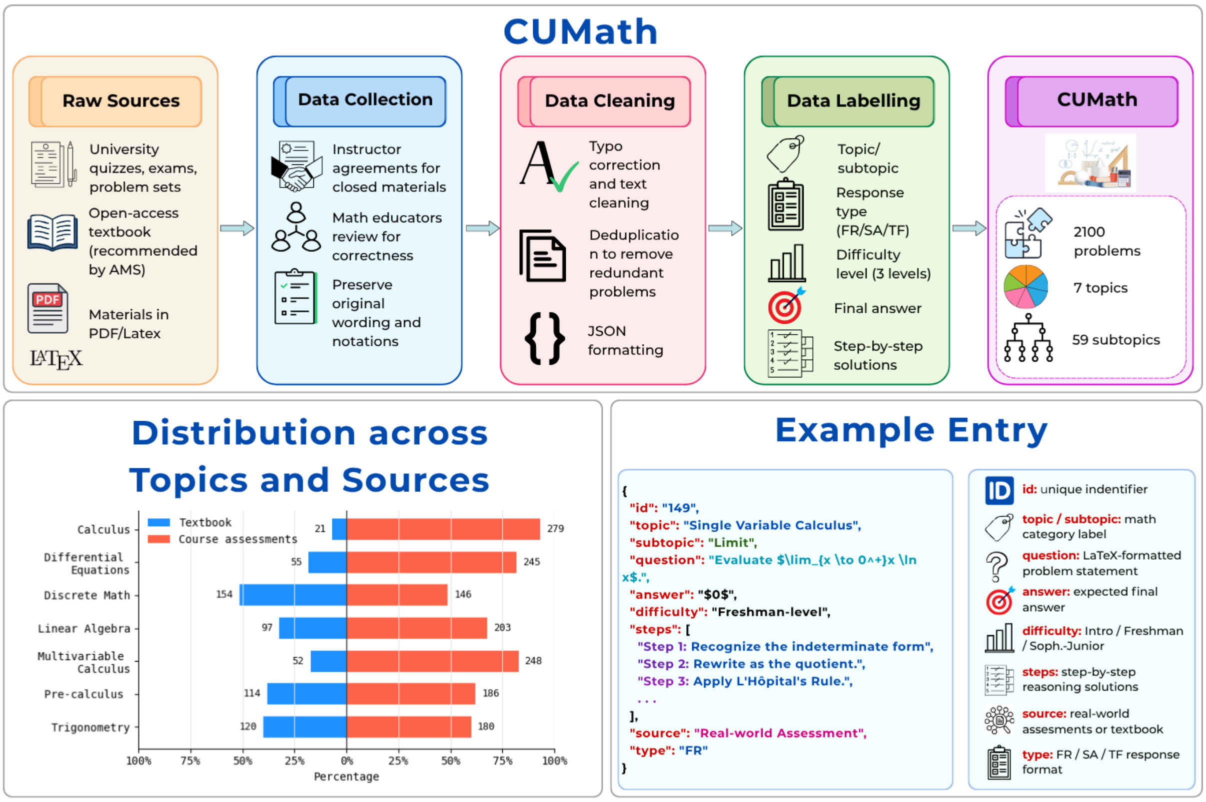

Figure 1: Overview of the CUMath dataset construction pipeline, topic distribution, and data example entry.

Hockenmaier, 2025). To operationalize these criteria, existing approaches employ various evaluators, including rule-based or symbolic checkers (Nguyen et al., 2024; Saparov and He, 2023), natural language inference (NLI) models (Golovneva et al., 2023), and LLM-as-value-function (Cui et al., 2025). Nevertheless, the reliability of these metrics for evaluating reasoning traces remains limited (Lee and Hockenmaier, 2025).

Human annotation has also been used to assess reasoning quality (Cobbe et al., 2021). These methods can provide reliable, high-quality judgments about the correctness of reasoning, but large-scale annotation of intermediate reasoning is costly and requires substantial domain expertise (Lightman et al., 2024). Thus, existing resources either focus on final-answer correctness or contain limited expert-verified intermediate steps.

3 CUMath Dataset

3.1 CUMath Overview

We present CUMath, a benchmark for evaluating undergraduate mathematical reasoning step-by-step. Unlike existing datasets that focus on artificial or competition-style problems, CUMath is collected from real undergraduate course materials, providing more authentic representations of the challenges students encounter.

The dataset contains 2,100 problems evenly distributed across seven core areas of undergraduate mathematics: Calculus, Differential Equations, Discrete Mathematics, Linear Algebra, Multivariable Calculus, Precalculus, and Trigonometry, with exactly 300 problems per subject. Unlike previous datasets that often overrepresented certain domains, this uniform distribution prevents topic-specific biases, enabling fairer comparisons of model performance and supporting a more comprehensive assessment of mathematical reasoning. Problems are categorized into three answer formats: Free Response (FR), Short Answer (SA), and True/False (TF). Each problem also includes detailed step-by-step solutions, enabling a comprehensive evaluation of understanding that extends beyond simply checking for the final answer's accuracy.

3.2 CUMath Creation

CUMath was constructed through 3 phases: **Data Collection**, **Data Cleaning**, and **Data Labeling**.

Data Collection. CUMath problems are collected from two primary sources: (i) [anonymized] university quizzes, exams, and problem sets, and (ii) open-access textbooks recommended by (American Institute of Mathematics) and released under Creative Commons licenses. Closed materials were included through instructors' agreements (Appendix A). Although we cannot guarantee exclusion from LLM training data, licensing restrictions and the private nature of course materials reduce this likelihood. All problems were reviewed by mathematics educators for clarity and correctness while preserving the original wording and notation.

Data Cleaning. Each problem was converted into a structured JSON format for consistent access and downstream use. Text was cleaned to correct typographical and formatting errors, while mathematical expressions were encoded in LaTeX for compatibility with language model inputs. We also performed deduplication to remove redundant problems and ensure that each item was self-contained.

Data Labeling. Each problem was annotated with metadata including a unique identifier, topic and subtopic labels, question text, source attribution, response format, final answer, and step-by-step solution to support both final-answer and intermediate-reasoning evaluation. Problems are categorized into three response types: FR, SA, and TF, reflecting common assessment styles in mathematics courses. We also assigned each problem a difficulty level based on the typical stage of undergraduate coursework in which the required concepts are taught: Introductory, Freshman-level, and Sophomore/Junior-level.

4 Evaluation Metrics

4.1 Automatic Metrics

To comprehensively evaluate final-answer correctness and step-by-step reasoning quality, we assess model performance by integrating four automatic metrics: Accuracy, Semantic F1, Stepwise Reasoning Score, and Validity–Redundancy Score.

Let $\mathcal{D} = \{(q_i, a_i, S_i)\}_{i=1}^{N}$ denote the dataset, where q_i is a problem, a_i its ground-truth answer, and S_i its reference reasoning steps. Given a model M, let $\hat{a}_i$ and $\hat{S}_i$ denote the predicted answer and reasoning steps.

Accuracy. We evaluate final-answer correctness using symbolic equivalence. Let $\phi(\cdot)$ denote a SymPy-based symbolic parser and $\mathbb{I}[\cdot]$ the indicator function. If symbolic parsing fails, exact string matching is used. Accuracy is computed as:

$$\text{Accuracy}(M) = \frac{1}{N}\sum_{i=1}^{N}\mathbb{I}[\phi(\hat{a}_i) \equiv \phi(a_i)]\,.$$

Semantic F1. To evaluate semantic alignment between generated and reference reasoning steps, we encode each step using a pretrained sentence encoder and compute pairwise cosine similarity. Following prior embedding-based matching methods (Zhang* et al., 2020; Mohamed and Oussalah, 2020), we construct a greedy one-to-one alignment using a similarity threshold of 0.7. Let M_i denote the number of matched step pairs. Dataset-level precision, recall, and F1 are computed as:

$$P = \frac{\sum_i M_i}{\sum_i |\hat{S}_i|}, \qquad R = \frac{\sum_i M_i}{\sum_i |S_i|},$$

$$\text{F1}(M) = \frac{2PR}{P+R}.$$

Stepwise Reasoning Score (SRS). Following ROSCOE (Golovneva et al., 2023), we assess reasoning quality using six normalized metrics: Faithfulness, Informativeness (Step), Informativeness (Chain), Coherence, Discourse Representation, and Repetition. For reasoning trace e_i, the overall score is:

$$\text{SRS}(e_i) = \frac{1}{6}\sum_{k=1}^{6} m_k(e_i),$$

and the final dataset score is averaged across all examples.

Validity–Redundancy (VR). We adapt ReasonEval (Xia et al., 2025), which scores each reasoning step using an NLI model. For step $\hat{s}_i^j$, validity is defined as the sum of entailment and neutral probabilities, while redundancy corresponds to the neutral probability. The final score is:

$$\text{VR}(e_i) = \min_j S_j^{\text{validity}} - \max_j S_j^{\text{redundancy}},$$

with dataset-level scores computed by averaging across examples.

4.2 LLM as a grader

We design an automatic grading pipeline that assesses both the correctness of final answers and the quality of the written solution (Figure 2). This matters because two solutions with the same final answer can come from very different reasoning.

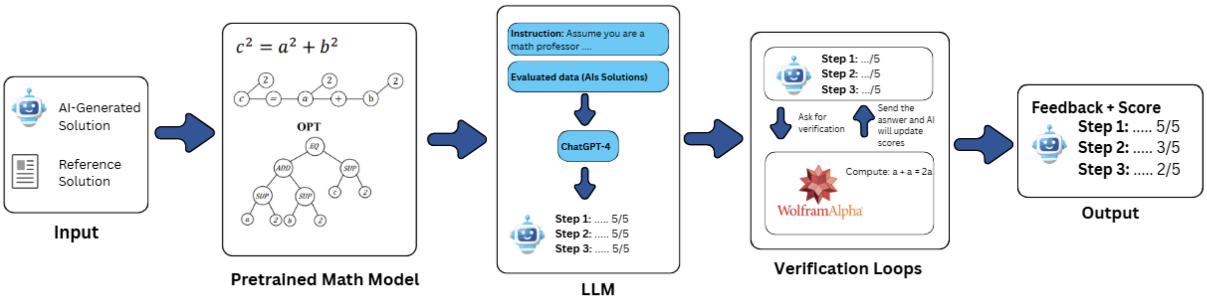

Figure 2: Overview of our grading pipeline. MathBERT encodes expressions, LLM gives step-level feedback, and an external Computer Algebra System (CAS) verifies correctness.

Our pipeline takes as input both AI-generated solutions and reference solutions, so each step can be compared against a trusted path.

Step 1 (Input). The pipeline begins with both AI-generated and reference solutions, which together provide a basis for comparison against a trusted path.

Step 2 (Math Segmentation). To process a student or AI-generated solution, we first perform a step segmentation procedure. Since mathematical solutions are written in free-form text, the pipeline needs a consistent way to isolate units of reasoning. If the solution is explicitly structured with steps labeled such as "step k", we use those as natural boundaries. In cases without explicit markers, we default to line-based segmentation, where each line is treated as a candidate reasoning step. Each extracted step is then encoded using MathBERT (Peng et al., 2021), which preserves the structure of equations, improving the accuracy in comparing generated and reference steps.

Step 3 (LLM Feedback). The encoded steps, along with a textual representation of their embeddings, are provided to an LLM prompted to act as a mathematical instructor (see Appendix E). The LLM uses this information to deliver step-level feedback by identifying errors, reasoning gaps, and partial correctness. In addition to qualitative comments, the LLM assigns a preliminary score on a 0–5 scale reflecting the validity and clarity of each step.

Step 4 (Verification Loops). To improve reliability, we integrate verification loops with external CAS (i.e, Wolfram Alpha). Whenever the pipeline detects that a reasoning step contains a mathematical expression, that expression is extracted, normalized, and sent as a query to the CAS. The CAS then returns the mathematically validated result, such as the simplified form of an equation, the solved solution set, or confirmation of equivalence between two expressions. The pipeline compares the LLM's judgment of the step with the CAS's authoritative output. If the CAS verifies the equivalence, the LLM's proposed assessment is maintained. If a discrepancy arises, for example, when the LLM accepts an invalid manipulation or fails to recognize an equivalence, the CAS result takes priority, and the LLM is prompted to revise its assessment based on the verified computation. This proposer–verifier loop reduces hallucinations, arithmetic mistakes, and symbolic misinterpretations, while ensuring that the grader remains consistent with formal mathematics.

Step 5 (Output). The final output is step-level feedback and a numerical score, combining the LLM's reasoning-based assessment with CAS verification to ensure mathematical validity.

4.3 Human Validation

We conducted human evaluations on subsets of the benchmark to validate both the automatic metrics and the LLM-as-a-grader to assess the reliability of our evaluation framework. Three annotators with sufficient mathematical background evaluated undergraduate-level multi-step reasoning solutions. Annotators independently assessed each solution without access to model identities or automatic metric scores. Because the automatic metrics and the LLM-as-a-grader evaluate reasoning differently, we used two different human-evaluation protocols.

4.3.1 Validation of Automatic Metrics

For the automatic metrics, annotators followed detailed evaluation guidelines that guided them in breaking down each solution and assessing the intermediate reasoning according to the same structured criteria used by the automatic evaluation pipeline. This provides a controlled comparison between human judgments and the step-level reasoning signals measured by SRS and VR.

We randomly sampled 5 problems from each of the 7 topics, resulting in 35 evaluation problems. For each problem, we collected solutions from four LLMs: GPT-5.2, together with the highest-performing open-source model, the highest-performing math-specialized model, and the highest-performing closed-source model excluding GPT-5.2 within each topic. All solutions were generated using Chain-of-Thought (CoT) prompting, which achieved the strongest overall performance across Accuracy, Semantic F1, SRS, and VR metrics (Section 7.2). In total, annotators evaluated 140 solutions. We measured inter-annotator agreement using Krippendorff's α and additionally treated the automatic metrics as additional annotators to measure alignment between automatic and human judgments.

4.3.2 Validation of LLM-as-a-Grader

We separately evaluated the reliability of the LLM-as-a-grader using a more free-form human evaluation protocol. In this setting, annotators did not explicitly decompose the solution into individual automatic metric components. Instead, they assigned a score from 1 to 5 for each reasoning step based on correctness, logical flow, and clarity.

We used the same sampling strategy of five problems from each of the seven topics. For each problem, we used solutions from three representative model families (closed-source, open-source, and math-specialized). For each topic, we selected the highest-performing model from each family with GPT-5.2 excluded. This resulted in 105 evaluated solutions. We first measured agreement among the three human annotators using Krippendorff's α and pairwise quadratic-weighted Cohen's κ. We then treated the LLM-as-a-grader as an additional annotator and computed the same agreement measures to evaluate how closely its judgments aligned with human evaluations.

5 Experiments

5.1 Models

Our evaluation covers 3 categories of LLMs to provide a comprehensive analysis of mathematical reasoning. Closed-source models demonstrate proprietary advancements, open-source models emphasize transparency and community collaboration, and math-specialized models are optimized for symbolic reasoning in targeted assessments. **Closed-source models** include GPT-5.2, GPT-4.1, OpenAI o3, Claude Sonnet 3.7. **Open-source models** include DeepSeek-R1-Distill-Qwen-32B, Gemma 2 9B IT, LLaMA 3 8B Instruct, LLaMA 3 70B Instruct, LLaMA 4 Scout 17B Instruct, Qwen2.5 7B Instruct, Mistral 7B Instruct v0.3. **Math-specialized models** include Qwen2.5-Math-7B Instruct, Qwen2.5-Math-1.5B Instruct, Llemma-7B, LLaMA-3.2-1B Instruct (ft). Detailed model specifications are provided in Appendix C.

5.2 Evaluation Strategies

We evaluate four prompting techniques commonly used to enhance reasoning in LLMs—Zero-shot, Chain-of-Thought (CoT), Self-Consistency (SC), and Tree-of-Thoughts (ToT)—to study their effects on final-answer accuracy and intermediate reasoning. The full set of prompt templates used in our experiments is provided in Appendix D.

To ensure consistency and reproducibility, we standardize decoding parameters across all models. Specifically, we employ greedy decoding with temperature set to 0 for both Zero-Shot and CoT prompting, as lower temperatures produce more deterministic outputs and prior work has shown that temperatures in the range 0.0–1.0 do not significantly affect problem-solving performance (Renze, 2024). For SC, we sample 5 reasoning chains at temperature 0.9 and select the final answer by majority vote, using the best-performing SC configuration (Wang et al., 2023b). For ToT, we generate 3 distinct reasoning paths at temperature 0.7, following the best-performing ToT configuration (Yao et al., 2023), to encourage exploratory reasoning.

All outputs are constrained to a maximum length of 2,048 tokens. This limit is sufficient to capture the complete reasoning process and final answers for all problems in our dataset, while fitting within the context window of all evaluated models. We evaluate model performance using 4 automatic metrics described in Section 4, namely Accuracy, Semantic F1, SRS, and VR.

6 Main Results

6.1 Overall Performance

Across the evaluated models, GPT-5.2 consistently outperforms the other LLMs in both final-answer accuracy and reasoning-oriented metrics.

As shown in Figure 3, GPT-5.2 achieves the highest Accuracy (0.293), Semantic F1 (0.330), and SRS (0.330). It also obtains the best average VR score (-0.121), substantially closer to zero than all other models, indicating fewer reasoning violations and less redundant intermediate reasoning. Even with these relative improvements, performance on CUMath remains substantially below the results reported on common mathematics benchmarks.

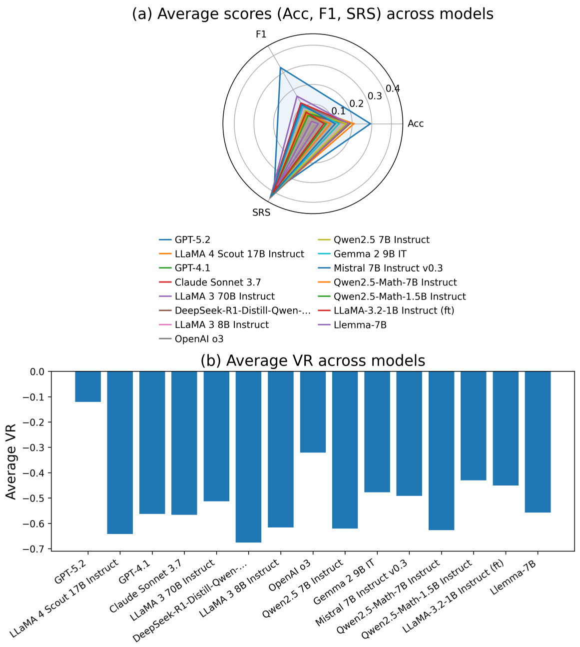

Figure 3: **Multi-metric summary of CUMath results.** (a) Average scores on Acc, F1, and SRS across models. (b) Average VR scores across models.

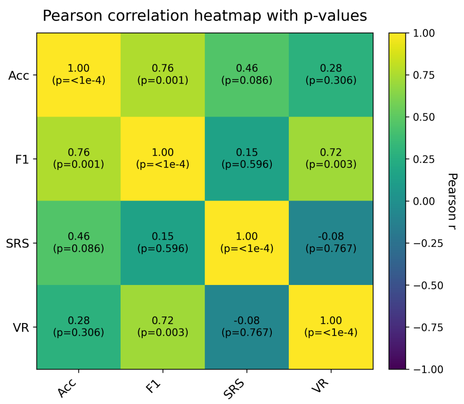

Figure 4: Pearson correlation heatmap among evaluation metrics (Accuracy, F1, SRS, and VR) with p-value

Despite recent claims that LLMs (e.g., GPT-5) are increasingly approaching expert-level reasoning capabilities (OpenAI, 2025), their performance on CUMath remains far below that expected of undergraduate students in the corresponding subject areas. Even when models arrive at correct final answers, their reasoning traces frequently diverge from human reference solutions and often fail to maintain logically valid intermediate reasoning. This behavior is reflected in the relatively low Semantic F1 and consistently negative VR scores across models, suggesting unreliable intermediate reasoning despite correct final answers.

More importantly, higher answer accuracy does not necessarily imply reliable reasoning. As shown in Figure 4, accuracy shows only a moderate and statistically non-significant correlation with SRS ($r = 0.46$, $p = 0.086$) and a weak correlation with VR ($r = 0.28$, $p = 0.306$). In contrast, F1 strongly correlates with Accuracy ($r = 0.76$, $p = 0.001$) and VR ($r = 0.72$, $p = 0.003$), suggesting that models with higher accuracy primarily produce reasoning traces that are semantically closer to human-written solutions. However, correct answers can still arise from flawed or weakly justified intermediate reasoning, while incorrect solutions often diverge substantially from human solution processes, as further examined through representative failure cases in Section 8.

These results suggest that benchmark accuracy alone may overestimate reasoning competence in modern LLMs. Since users increasingly interact with generated reasoning traces for trust calibration and solution verification (Chen et al., 2026), the reliability of intermediate reasoning becomes as important as final-answer correctness. This poses an important challenge for current LLMs: models must provide not only correct answers but also reliable, logically consistent reasoning traces. However, our step-level evaluation shows that current LLMs frequently fail to maintain logically grounded reasoning across solution chains.

6.2 LLM-as-a-Grader Evaluation

We evaluate model-generated solutions using our LLM-as-a-grader pipeline (Figure 5). GPT-5.2 is excluded because it was not evaluated by the LLM-as-a-grader.

Across models, the LLM-as-a-grader generally assigns higher scores to stronger closed-source and larger open-source models. Claude Sonnet 3.7 achieves the highest average score ($3.40/5$), while math-specialized models receive substantially lower scores (e.g., $0.89/5$ for Llemma-7B). This gap suggests that model differences extend beyond final-answer correctness: stronger models tend to produce intermediate reasoning that is more logically coherent, better justified, and clearer. At the same time, the variation in grader scores across models provides an additional signal of reasoning quality that is not captured by final-answer accuracy or any individual automatic metric.

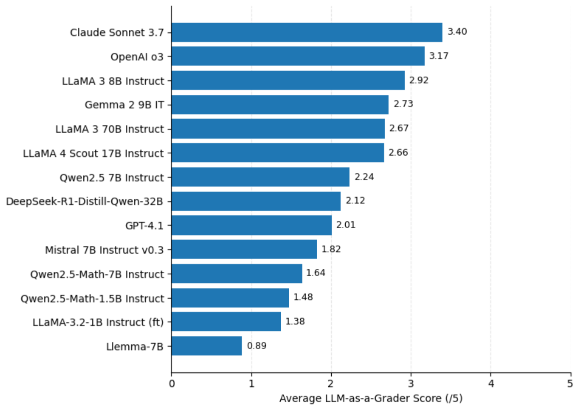

Figure 5: LLM-as-a-grader evaluation across models.

6.3 Human Agreement and Evaluation Reliability

6.3.1 Automatic Metric Reliability

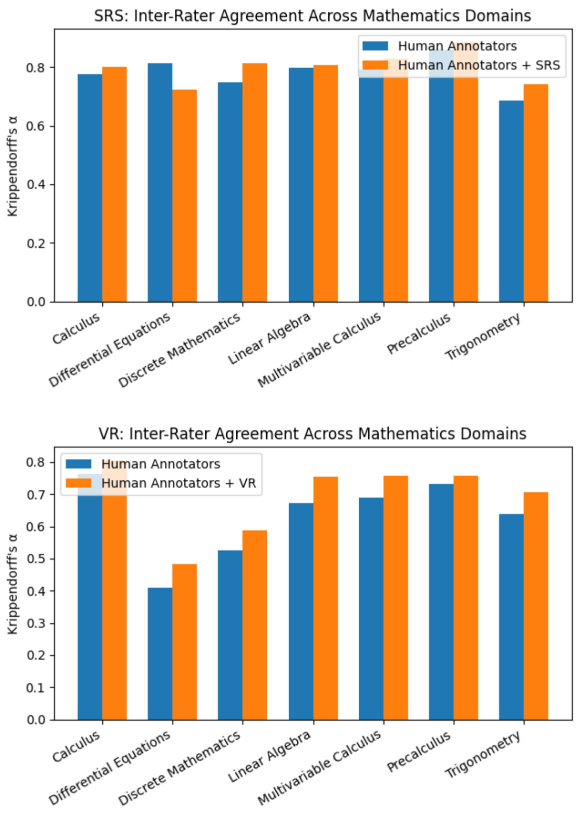

Figure 6: Krippendorff's α agreement scores across mathematics domains for SRS and VR.

Figure 6 shows that the automatic metrics are reasonably aligned with human judgments. Both metrics (e.g, SRS and VR) demonstrated substantial agreement with human annotators across most topics. Lower agreement was observed in Differential Equations and Discrete Mathematics, which these domains often involve more open-ended reasoning and multiple valid solution strategies. The agreement scores remain relatively consistent across domains for both SRS and VR, indicating that the automatic metrics arguably induce grading behavior that remains reasonably close to that of human annotators under structured evaluation criteria.

6.3.2 LLM-as-a-Grader Reliability

To evaluate the reliability of our human evaluation setup and the alignment of the LLM-as-a-grader with human graders, we report inter-rater agreement across human–human and human–LLM comparisons (Figure 7).

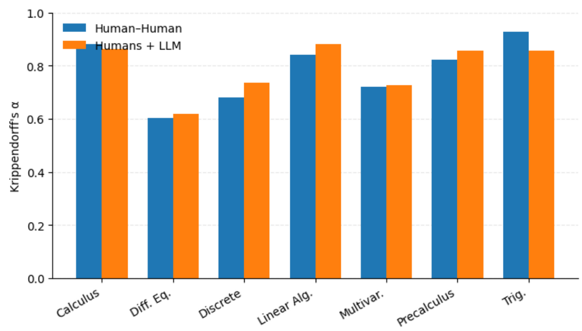

Figure 7: Krippendorff's α for human–human and human–LLM comparisons across topics.

Across all solutions, the three human annotators achieve an overall Krippendorff's α of 0.829, indicating strong reliability (Figure 7). Topic-level agreement is consistently high, with α values ranging from 0.60 to 0.93 across domains. Pairwise Cohen's κ follows the same pattern, ranging between 0.42–0.93 (Figure 8).

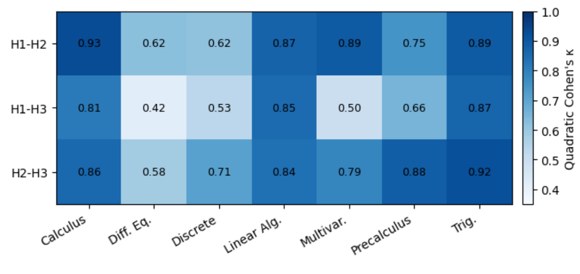

Figure 8: Human–human quadratic Cohen's κ across topics.

We evaluate the alignment between human annotators and the LLM using the same reliability metrics (Figures 7 and 9). The overall Krippendorff's α remains nearly unchanged at 0.832, indicating that the LLM preserves dataset-level reliability and aligns well with human judgments. Topic-level agreement ranges from 0.62 to 0.88, with κ values comparable to human–human variability. These results suggest that the LLM-as-a-grader produces judgments that are broadly consistent with human evaluations and remain within the variability observed among human annotators.

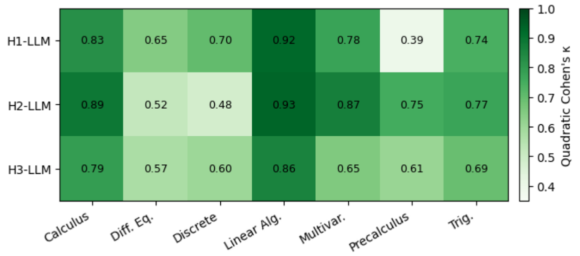

Figure 9: Human–LLM quadratic Cohen's κ across topics.

7 Implications for Reasoning Evaluation

7.1 Undergraduate Mathematics as a Benchmark for Reasoning

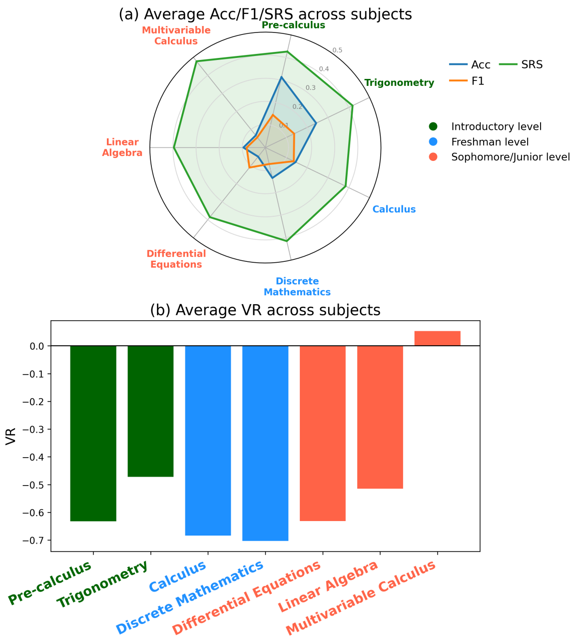

Figure 10: Average Acc, F1, SRS, VR by subjects.

Our results highlight that strong final-answer accuracy in lower-level mathematics does not necessarily translate into reliable reasoning on problems, especially the ones that require sustained symbolic manipulation and multi-step derivations. Figure 10 shows models perform noticeably worse on sophomore- and junior-level subjects (e.g., Differential Equations, Multivariable Calculus, and Linear Algebra) than on introductory topics (e.g., Trigonometry and Precalculus). Subjects requiring more abstract or multi-step reasoning, such as Differential Equations, Linear Algebra, and Discrete Mathematics, also exhibit lower Semantic F1 and VR scores, suggesting weaker alignment with human-written reasoning processes and less reliable intermediate reasoning.

These findings indicate that outcome-based evaluation alone may overestimate reasoning reliability, as models can still produce mathematically plausible explanations that diverge from standard human solution procedures. Such behaviors are particularly concerning in educational and human-AI problem-solving settings, where users may rely on intermediate reasoning steps for verification and trust calibration. Representative failure cases are further discussed in Section 8.

7.2 Prompting and Process-Level Reasoning

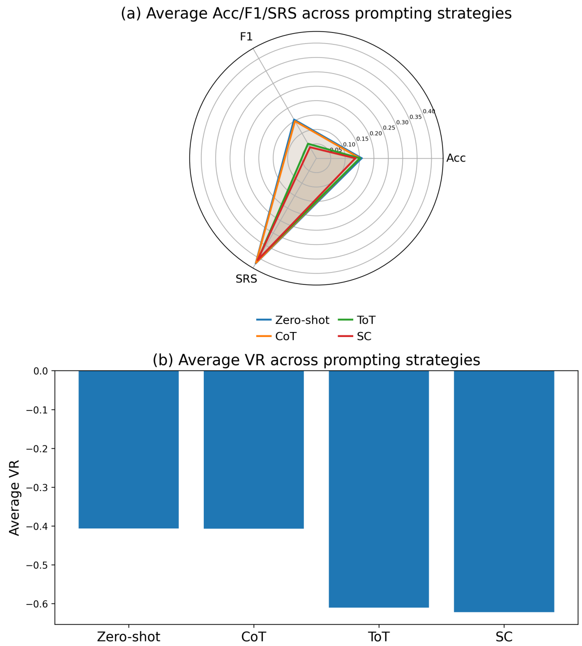

Figure 11: Average Acc, F1, SRS, VR by prompting.

Different prompting strategies substantially affect reasoning quality, consistent with prior findings on prompt sensitivity in LLMs (Zhuo et al., 2024). As shown in Figure 11, CoT achieves the strongest overall performance in Accuracy, Semantic F1, and

SRS across model categories, while obtaining VR scores comparable to Zero-shot prompting. However, CoT provides only limited improvement over Zero-shot prompting, suggesting that generating step-by-step explanations alone is insufficient to improve process-level reasoning reliability.

Similarly, ToT and SC do not substantially improve accuracy and consistently reduce F1, SRS, and VR scores. These results suggest that prompting strategies encouraging broader exploration or multiple reasoning paths do not necessarily improve process-level reasoning quality. Although ToT and SC generate more diverse reasoning traces, these solutions frequently diverge from standard human solution procedures and often lack sufficient logical support for the final answer. These findings suggest that prompting strategies primarily influence the structure and style of generated reasoning rather than substantially improving reasoning reliability. Future work should therefore focus less on generating longer reasoning traces and more on improving the consistency and verifiability of intermediate reasoning.

8 A Case Study

Despite recent advances in LLMs, our analysis shows that they continue to make basic yet systematic errors on undergraduate-level mathematics. An example is the following pair of integrals:

$$\int \frac{1-\sin x}{x+\cos x}\,dx \qquad \text{and} \qquad \int_{-\pi/6}^{\pi/6} \frac{1-\sin x}{x+\cos x}\,dx.$$

Most students who have studied introductory calculus would immediately recognize that the two problems are closely related. In particular, once the indefinite integral is solved, the definite integral follows directly by evaluating the same antiderivative at the endpoints. Using the substitution $u = x+\cos x$, we have $du = (1-\sin x)\,dx$, giving

$$\int \frac{1-\sin x}{x+\cos x}\,dx = \ln|x+\cos x| + C,$$

and therefore

$$\int_{-\pi/6}^{\pi/6} \frac{1-\sin x}{x+\cos x}\,dx = [\ln|x+\cos x|]_{-\pi/6}^{\pi/6} \approx 1.4.$$

The model responses did not show the same consistency. When asked for the *indefinite integral*, most models correctly applied the substitution and obtained the correct antiderivative. However, when asked for the corresponding *definite integral*, many models failed to reuse the antiderivative and instead switched to unrelated arguments or incorrect shortcuts. For example, **GPT-4.1** abandoned the substitution and incorrectly argued that the integrand is odd, concluding that the integral over the symmetric interval $[-\pi/6, \pi/6]$ must be zero (see Solution 2 in Appendix H). This reasoning is invalid, since $f(x) = (1-\sin x)/(x+\cos x)$ is not an odd function. Interestingly, the model had already demonstrated knowledge of the correct antiderivative in the indefinite case, suggesting a breakdown in reasoning consistency rather than a failure to identify the appropriate substitution.

A different failure mode appears in models that produced values close to the correct result through incorrect reasoning. For example, **OpenAI-o3** and **Mistral 7B Instruct** returned values close to the true answer only after incorrectly claiming that no closed-form antiderivative exists and switching to numerical approximation (see Solution 4 and Solution 5 in Appendix H). Although the numerical results were close to the correct value, the underlying reasoning process remained flawed.

A different type of reasoning behavior appears in **GPT-5.2**. Unlike OpenAI-o3 and Mistral 7B Instruct, which switched directly to numerical approximation, GPT-5.2 pursued a substantially longer and more non-standard symbolic derivation before eventually returning to the correct result (see Solution 3 in Appendix H). Rather than directly reusing the substitution from the indefinite integral case, the model introduced additional symmetry arguments and intermediate transformations, unnecessarily complicating the solution process compared to standard human solution procedures.

These examples illustrate several recurring weaknesses in LLM mathematical reasoning, including inconsistent symbolic reasoning, reliance on unjustified shortcuts, and unnecessarily complicated reasoning paths that diverge from standard human approaches. Importantly, many of these solutions can still appear coherent and mathematically convincing despite containing flawed or inefficient intermediate reasoning. This poses a challenge for educational and human–AI problem-solving settings, where users may rely on intermediate reasoning steps for verification and trust calibration. As a result, evaluations based solely on final-answer accuracy may substantially overestimate the true reasoning reliability of current LLMs.

9 Conclusion

We introduced CUMath, a benchmark for evaluating undergraduate-level mathematical reasoning with detailed step-by-step solutions. Using a multi-metric evaluation framework, we evaluated 15 LLMs across multiple prompting strategies and analyzed both final-answer correctness and intermediate reasoning quality. Our results show that final-answer accuracy alone is insufficient to evaluate reasoning reliability, and that undergraduate mathematics remains a challenging benchmark for evaluating LLMs' performance and process-level reasoning. These findings highlight the importance of moving beyond outcome-based evaluation toward assessing the validity and consistency of intermediate reasoning, particularly in educational and human-AI problem-solving settings.

Limitations

Although CUMath provides a reasoning-focused benchmark for undergraduate mathematics, our work has several limitations. First, our evaluation does not include some of the most recently released LLMs, and future evaluations using CUMath should incorporate newer models to reflect evolving capabilities. Second, step-by-step evaluation relies on predefined automatic metrics, which may not fully capture undergraduate mathematical validity; although we conducted human validation to assess metric reliability, automatic evaluation still cannot fully replace expert mathematical judgment. Third, step-level analysis depends on heuristic segmentation of model-generated solutions (e.g., by newlines or explicit step markers), which may not align with the true logical structure of reasoning, and developing more principled segmentation methods remains an important direction for future work.

Ethical Consideration

CUMath is constructed from real instructional materials (university quizzes, exams, problem sets) and open-access textbooks (Appendix A). Textbook items are released under their respective Creative Commons licenses; closed instructional materials were included under explicit agreements with instructors. The dataset contains no personally identifiable information, human-subject data, or sensitive attributes; problems were reviewed by mathematics educators for clarity and curricular alignment.

Potential risks include (i) inadvertent training set overlap with future models and (ii) misuse of the benchmark for assessing general reasoning ability. To mitigate these risks, we (a) release provenance metadata and licensing information and (b) distribute CUMath for non-commercial research use. We comply with the ACL Code of Ethics and the legal terms of all sources, and we utilize Grammarly to enhance the paper's grammar and clarity.

References

Janice Ahn, Rishu Verma, Renze Lou, Di Liu, Rui Zhang, and Wenpeng Yin. 2024. Large language models for mathematical reasoning: Progresses and challenges. *arXiv preprint arXiv:2402.00157*.

American Institute of Mathematics. Aim approved textbooks. https://aimath.org/textbooks/approved-textbooks/. Accessed: July 2025.

Johan Boye and Birger Moell. 2025. Large language models and mathematical reasoning failures. *arXiv preprint arXiv:2502.11574*.

Shiwei Chen, Niruthikka Sritharan, Xiaolin Wen, Chenxi Zhang, Xingbo Wang, and Yong Wang. 2026. When the chain breaks: Interactive diagnosis of llm chain-of-thought reasoning errors. *arXiv preprint arXiv:2603.21286*.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*.

Ganqu Cui, Lifan Yuan, Zefan Wang, Hanbin Wang, Wendi Li, Bingxiang He, Yuchen Fan, Tianyu Yu, Qixin Xu, Weize Chen, Jiarui Yuan, Huayu Chen, Kaiyan Zhang, Xingtai Lv, Shuo Wang, Yuan Yao, Xu Han, Hao Peng, Yu Cheng, and 4 others. 2025. Process reinforcement through implicit rewards. *CoRR*, abs/2502.01456.

Chunyuan Deng, Yilun Zhao, Xiangru Tang, Mark Gerstein, and Arman Cohan. 2024. Investigating data contamination in modern benchmarks for large language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 8706–8719, Mexico City, Mexico. Association for Computational Linguistics.

Meng Fang, Xiangpeng Wan, Fei Lu, Fei Xing, and Kai Zou. 2024. Mathodyssey: Benchmarking mathematical problem-solving skills in large language models using odyssey math data. *arXiv preprint arXiv:2406.18321*.

Sebastian Farquhar, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. 2024. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630(8017):625–630.

Olga Golovneva, Moya Chen, Spencer Poff, Martin Corredor, Luke Zettlemoyer, Maryam Fazel-Zarandi, and Asli Celikyilmaz. 2023. Roscoe: A suite of metrics for scoring step-by-step reasoning. *Preprint*, arXiv:2212.07919.

Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. 2021a. Measuring massive multitask language understanding. In *International Conference on Learning Representations*.

Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. 2021b. Measuring mathematical problem solving with the math dataset. *NeurIPS*.

Thomas Hubert, Rishi Mehta, Laurent Sartran, Miklós Z Horváth, Goran Žužić, Eric Wieser, Aja Huang, Julian Schrittwieser, Yannick Schroecker, Hussain Masoom, and 1 others. 2025. Olympiad-level formal mathematical reasoning with reinforcement learning. *Nature*, pages 1–3.

Jinu Lee and Julia Hockenmaier. 2025. Evaluating step-by-step reasoning traces: A survey. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 1789–1814, Suzhou, China. Association for Computational Linguistics.

Aitor Lewkowycz, Anders Johan Andreassen, David Dohan, Ethan Dyer, Henryk Michalewski, Vinay Venkatesh Ramasesh, Ambrose Slone, Cem Anil, Imanol Schlag, Theo Gutman-Solo, Yuhuai Wu, Behnam Neyshabur, Guy Gur-Ari, and Vedant Misra. 2022. Solving quantitative reasoning problems with language models. In *Advances in Neural Information Processing Systems*.

Xiaoyuan Li, Wenjie Wang, Moxin Li, Junrong Guo, Yang Zhang, and Fuli Feng. 2024. Evaluating mathematical reasoning of large language models: A focus on error identification and correction. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 11316–11360, Bangkok, Thailand. Association for Computational Linguistics.

Hunter Lightman, Vineet Kosaraju, Yuri Burda, Harrison Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. 2024. Let's verify step by step. In *The Twelfth International Conference on Learning Representations*.

Bang Liu, Xinfeng Li, Jiayi Zhang, Jinlin Wang, Tanjin He, Sirui Hong, Hongzhang Liu, Shaokun Zhang, Kaitao Song, Kunlun Zhu, Yuheng Cheng, Suyuchen Wang, Xiaoqiang Wang, Yuyu Luo, Haibo Jin, Peiyan Zhang, Ollie Liu, Jiaqi Chen, Huan Zhang, and 28 others. 2025. Advances and challenges in foundation agents: From brain-inspired intelligence to evolutionary, collaborative, and safe systems. *Preprint*, arXiv:2504.01990.

Pan Lu, Hritik Bansal, Tony Xia, Jiacheng Liu, Chunyuan Li, Hannaneh Hajishirzi, Hao Cheng, Kai-Wei Chang, Michel Galley, and Jianfeng Gao. 2024. Mathvista: Evaluating mathematical reasoning of foundation models in visual contexts. *Preprint*, arXiv:2310.02255.

Muhidin Mohamed and Mourad Oussalah. 2020. A hybrid approach for paraphrase identification based on knowledge-enriched semantic heuristics. *Language Resources and Evaluation*, 54(2):457–485.

Minh-Vuong Nguyen, Linhao Luo, Fatemeh Shiri, Dinh Phung, Yuan-Fang Li, Thuy-Trang Vu, and Gholamreza Haffari. 2024. Direct evaluation of chain-of-thought in multi-hop reasoning with knowledge graphs. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 2862–2883, Bangkok, Thailand. Association for Computational Linguistics.

OpenAI. 2025. Introducing gpt-5. `https://openai.com/index/introducing-gpt-5/`.

Shuai Peng, Ke Yuan, Liangcai Gao, and Zhi Tang. 2021. Mathbert: A pre-trained model for mathematical formula understanding. *Preprint*, arXiv:2105.00377.

Runqi Qiao, Qiuna Tan, Guanting Dong, Minhui Wu, Chong Sun, Xiaoshuai Song, Zhuoma GongQue, Shanglin Lei, Zhe Wei, Miaoxuan Zhang, Runfeng Qiao, Yifan Zhang, Xiao Zong, Yida Xu, Muxi Diao, Zhimin Bao, Chen Li, and Honggang Zhang. 2024. We-math: Does your large multimodal model achieve human-like mathematical reasoning? *Preprint*, arXiv:2407.01284.

Matthew Renze. 2024. The effect of sampling temperature on problem solving in large language models. In *Findings of the association for computational linguistics: EMNLP 2024*, pages 7346–7356.

Abulhair Saparov and He He. 2023. Language models are greedy reasoners: A systematic formal analysis of chain-of-thought. In *The Eleventh International Conference on Learning Representations*.

Parshin Shojaee, Iman Mirzadeh, Keivan Alizadeh, Maxwell Horton, Samy Bengio, and Mehrdad Farajtabar. 2025. The illusion of thinking: Understanding the strengths and limitations of reasoning models via the lens of problem complexity. *Preprint*, arXiv:2506.06941.

Charlie Snell, Jaehoon Lee, Kelvin Xu, and Aviral Kumar. 2024. Scaling llm test-time compute optimally can be more effective than scaling model parameters. *arXiv preprint arXiv:2408.03314*.

Mingyang Song, Zhaochen Su, Xiaoye Qu, Jiawei Zhou, and Yu Cheng. 2025. PRMBench: A fine-grained and challenging benchmark for process-level reward models. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 25299–25346, Vienna, Austria. Association for Computational Linguistics.

Huaijie Wang, Shibo Hao, Hanze Dong, Shenao Zhang, Yilin Bao, Ziran Yang, and Yi Wu. 2025a. Offline reinforcement learning for LLM multi-step reasoning. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 8881–8893, Vienna, Austria. Association for Computational Linguistics.

Lei Wang, Wanyu Xu, Yihuai Lan, Zhiqiang Hu, Yunshi Lan, Roy Ka-Wei Lee, and Ee-Peng Lim. 2023a. Plan-and-solve prompting: Improving zero-shot chain-of-thought reasoning by large language models. *arXiv preprint arXiv:2305.04091*.

Peng-Yuan Wang, Tian-Shuo Liu, Chenyang Wang, Ziniu Li, Yidi Wang, Shu Yan, Chengxing Jia, Xu-Hui Liu, Xinwei Chen, Jiacheng Xu, and 1 others. 2025b. A survey on large language models for mathematical reasoning. *ACM Computing Surveys*.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023b. Self-consistency improves chain of thought reasoning in language models. *Preprint*, arXiv:2203.11171.

Yiping Wang, Qing Yang, Zhiyuan Zeng, Liliang Ren, Liyuan Liu, Baolin Peng, Hao Cheng, Xuehai He, Kuan Wang, Jianfeng Gao, and 1 others. 2025c. Reinforcement learning for reasoning in large language models with one training example. *arXiv preprint arXiv:2504.20571*.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, and 1 others. 2022. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems*, 35:24824–24837.

Yangzhen Wu, Zhiqing Sun, Shanda Li, Sean Welleck, and Yiming Yang. 2024. Inference scaling laws: An empirical analysis of compute-optimal inference for problem-solving with language models. *arXiv preprint arXiv:2408.00724*.

Shijie Xia, Xuefeng Li, Yixin Liu, Tongshuang Wu, and Pengfei Liu. 2025. Evaluating mathematical reasoning beyond accuracy. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 27723–27730.

Xin Xu, Jiaxin Zhang, Tianhao Chen, Zitong Chao, Jishan Hu, and Can Yang. 2025. Ugmathbench: A diverse and dynamic benchmark for undergraduate-level mathematical reasoning with large language models. *arXiv preprint arXiv:2501.13766*.

Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas L. Griffiths, Yuan Cao, and Karthik Narasimhan. 2023. Tree of thoughts: Deliberate problem solving with large language models. *Preprint*, arXiv:2305.10601.

Xiang Yue, Yuansheng Ni, Kai Zhang, Tianyu Zheng, Ruoqi Liu, Ge Zhang, Samuel Stevens, Dongfu Jiang, Weiming Ren, Yuxuan Sun, Cong Wei, Botao Yu, Ruibin Yuan, Renliang Sun, Ming Yin, Boyuan Zheng, Zhenzhu Yang, Yibo Liu, Wenhao Huang, and 3 others. 2024. Mmmu: A massive multi-discipline multimodal understanding and reasoning benchmark for expert agi. In *Proceedings of CVPR*.

Renrui Zhang, Dongzhi Jiang, Yichi Zhang, Haokun Lin, Ziyu Guo, Pengshuo Qiu, Aojun Zhou, Pan Lu, Kai-Wei Chang, Peng Gao, and Hongsheng Li. 2024. Mathverse: Does your multi-modal llm truly see the diagrams in visual math problems? *Preprint*, arXiv:2403.14624.

Tianyi Zhang*, Varsha Kishore*, Felix Wu*, Kilian Q. Weinberger, and Yoav Artzi. 2020. Bertscore: Evaluating text generation with bert. In *International Conference on Learning Representations*.

Kunhao Zheng, Jesse Michael Han, and Stanislas Polu. 2022. Minif2f: a cross-system benchmark for formal olympiad-level mathematics. *Preprint*, arXiv:2109.00110.

Aojun Zhou, Ke Wang, Zimu Lu, Weikang Shi, Sichun Luo, Zipeng Qin, Shaoqing Lu, Anya Jia, Linqi Song, Mingjie Zhan, and Hongsheng Li. 2023. Solving challenging math word problems using gpt-4 code interpreter with code-based self-verification. *Preprint*, arXiv:2308.07921.

Jingming Zhuo, Songyang Zhang, Xinyu Fang, Haodong Duan, Dahua Lin, and Kai Chen. 2024. ProSA: Assessing and understanding the prompt sensitivity of LLMs. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 1950–1976, Miami, Florida, USA. Association for Computational Linguistics.

A Dataset Sources

CUMath also includes problems drawn from university-level quizzes and examinations authored by the course instructor. These materials are not publicly available; however, explicit licenses were obtained from the authors to incorporate them into our benchmark. We therefore designate these as *licensed instructor-authored problems*. Such items are tagged in the dataset metadata and are distributed only for non-commercial purposes.

Domain	Textbook Source	Author(s)	License
Calculus	*APEX Calculus*	Gregory Hartman	CC BY 4.0
Differential Equations	*Elementary Differential Equations (with BVP)*	William F. Trench	CC BY-SA 4.0
Discrete Mathematics	*Discrete Mathematics: An Open Introduction*	Oscar Levin	CC BY-SA 4.0
Linear Algebra	*A First Course in Linear Algebra*	Rob Beezer	CC BY-SA 4.0
Multivariable Calculus	*APEX Calculus*	Gregory Hartman	CC BY 4.0
Pre-calculus	*Precalculus / College Algebra / Trigonometry*	Carl Stitz, Jeff Zeager	CC BY-NC-SA 3.0
Trigonometry	*Precalculus / College Algebra / Trigonometry*	Carl Stitz, Jeff Zeager	CC BY-NC-SA 3.0

Table 2: Mapping of dataset domains to textbook sources and associated licenses.

B Sub-topic Distribution

Table 3: Distribution of problems across sub-topics in **Calculus**.

Sub-topic	Count
Definite Integral	53
Limit	52
Derivative	44
Indefinite Integral	42
Function Analysis	30
Sequence/Series	30
Real-world Problems (Optimization)	21
Continuity	17
Improper Integral	11

Table 4: Distribution of problems across sub-topics in **Differential Equations**.

Sub-topic	Count
Linear Second Order Equations	135
Laplace Transform	56
Linear First Order Equations	37
Exact Equations	22
Separable Equations	17
Transformation of Nonlinear Equations into Separable Equations	20
Existence and Uniqueness of Solutions of Nonlinear Equations	13

Table 5: Distribution of problems across sub-topics in **Discrete Mathematics**.

Sub-topic	Count
Sequences	106
Recurrences	81
Generating Functions	34
Number Theory	32
Sums & Products	27
Combinatorics	18
Logic	2

Table 6: Distribution of problems across sub-topics in **Linear Algebra**.

Sub-topic	Count
Linear Transformations	45
Linear Independence / Dependence	44
Eigenvalues, Eigenvectors & Characteristic Polynomial	36
Matrix Operations	34
Systems of Linear Equations	31
Spanning Sets, Rank & Dimension	27
Determinants	23
Matrix Properties & Operations	14
Vector Spaces / Subspaces	14
Vector Operations & Representations	10
Linear Transformations & Representations	9
Orthogonality / Inner Product	8
Null Space & Nullity	5

Table 7: Distribution of problems across sub-topics in **Multivariable Calculus**.

Sub-topic	Count
Vector Calculus	80
Multiple Integrals	74
Geometry of Space	49
Partial Derivatives	46
Limit	20
Function Analysis	19
Vector-valued Function	6
Real-world Problem (Optimization)	6

Table 8: Distribution of problems across sub-topics in **Pre-calculus**.

Sub-topic	Count
Functions	162
Applications	46
Equations	34
Polynomials	33
Log/Exponential	13
Inequalities	12

Table 9: Distribution of problems across sub-topics in **Trigonometry**.

Sub-topic	Count
Evaluating Trigonometric Functions	70
Inverse Trigonometric Functions	62
Trigonometric Equations	49
Exact Values	39
Trigonometric Identities	30
Solving Triangles	19
Angle Conversion	16
Unit Circle & Reference Angles	12
Trigonometric Inequalities	3

C Evaluated LLMs

Model	Type	Size	Release Date	Specialization
GPT-5.2	Closed-source	N/A	2025	General
GPT-4.1	Closed-source	N/A	2025	General
OpenAI o3	Closed-source	N/A	2024	General
Claude Sonnet 3.7	Closed-source	N/A	2025	General
DeepSeek-R1-Distill-Qwen-32B	Open-source	32B	2025	General
Gemma 2 9B IT	Open-source	9B	2024	General
LLaMA 3 8B Instruct	Open-source	8B	2024	General
LLaMA 3 70B Instruct	Open-source	70B	2024	General
LLaMA 4 Scout 17B Instruct	Open-source	17B	2025	General
Qwen2.5 7B Instruct	Open-source	7B	2025	General
Mistral 7B Instruct v0.3	Open-source	7B	2024	General
Qwen2.5-Math-7B Instruct	Math-specialized	7B	2024	Mathematical reasoning
Qwen2.5-Math-1.5B Instruct	Math-specialized	1.5B	2024	Mathematical reasoning
Llemma-7B	Math-specialized	7B	2024	Mathematical reasoning
LLaMA-3.2-1B Instruct (ai-nexuz ft.)	Math-specialized	1B	2024	Mathematical reasoning

Table 10: Detailed specifications of evaluated LLMs.

D Solution Generation Prompts

Chain-of-Thoughts Prompt

System Prompt: You are a highly skilled mathematics expert. Solve the problem step by step. Conclude with your final answer in the form:
`\boxed{your final answer here}`.
User: Q: {example-question-1}
A: {example-solution-steps}
Q: {example-question-2}
A: {example-solution-steps}
⋮
Q: {problem}
A:

Zero-Shot Prompt

System Prompt: Conclude the final answer in the form:
`\boxed{your final answer here}`.
User: Solve the following math problem:
{problem}

Tree-of-Thoughts Prompt

System: You are a highly skilled mathematics expert. Solve the problem using a Tree-of-Thoughts style approach: Propose exactly{TOT_PATHS} distinct intermediate solution paths. Select the most promising path. Continue developing only the selected path, revising if needed. Arrive at a final verified solution. At the end, state the final answer in the form: `\boxed{your final answer here}`. {problem}

Self-Consistency Prompt

System: You are a highly skilled mathematics expert. Solve the problem with clear, step-by-step reasoning. At the end, clearly state the final answer in the form: `\boxed{your final answer here}`.
{problem}
A (Sample 1): {reasoning}
A (Sample 2): {reasoning}
...
`\boxed{\{final-answer\}}`

E Evaluation Prompts (LLM-as-a-Grader)

Pass 1: Step Feedback + Score (No CAS)

System Prompt: You are a meticulous and fair mathematics instructor.
Given a problem, its correct reference steps, and a proposed step-by-step solution, evaluate each proposed step *independently*.
Score each step on a 1–5 scale (1 = very poor, 5 = excellent) based on: Correctness, Logic/Flow, Justification, and Clarity.
Important:

- Do *not* reference or claim any CAS results in this pass.
- If a step is prose (no explicit equality), still give feedback and a score.
- Judge each step as written; do not merge or rewrite steps.

Problem: {problem}
Reference Solution Steps:
Ref Step 1: {ref_step_1}
Ref Step 2: {ref_step_2}
⋮
Proposed Solution Steps:
Step 1: {model_step_1}
Step 2: {model_step_2}
⋮
Respond exactly in this format (one line per proposed step):
Step 1: [1–2 sentences of feedback] Score: [X]/5
Step 2: [1–2 sentences of feedback] Score: [Y]/5
⋮

Pass 2: CAS-Based Score Revision

System Prompt: You revise scores for mathematical reasoning steps using CAS results.
If CAS shows an incorrect transformation, lower the score; if it confirms the transformation, consider raising the score.
If both CAS statuses are unknown, keep the score unchanged and note `CAS unknown`.
Return a strict JSON list in the following format:
`[{"idx": int, "revised": int (1..5), "note": str}]`

F Detailed Computation of Stepwise Reasoning Score

We follow the ROSCOE framework (Golovneva et al., 2023) to evaluate the quality of reasoning chains produced by M. This section provides the exact computation of the six metrics we use: Faithfulness, Informativeness (Step), Informativeness (Chain), Repetition (Step), Discourse Representation, and Coherence (Step vs. Step). We implement faithfulness/informativeness via token/sentence–step cosine alignment and use an NLI model to penalize contradictions for Discourse/Coherence.

We represent each problem statement q_i as a sequence of tokens: $q_i = \{q_{i,1}, q_{i,2}, \ldots, q_{i,|q_i|}\}$, where $q_{i,t}$ denotes the embedding of the t-th token in q_i.

Faithfulness $(e_i \to q_i)$. Measures whether each generated step is grounded in the problem statement:

$$\text{Faithfulness}(e_i) = \frac{1}{\hat{n}_i}\sum_{j=1}^{\hat{n}_i} r\text{-align}(e_i^j \to q_i),$$

$$r\text{-align}(e_i^j \to q_i) = \frac{1 + \max_{t=1..|q_i|}\cos(e_i^j, q_{i,t})}{2}.$$

Informativeness (Step) $(e_i \leftrightarrow q_i)$. Captures how well information in the problem statement is reflected in the generated reasoning:

$$\text{Info-Step}(e_i) = \frac{1}{2}\left(\frac{1}{|q_i|}\sum_{t=1}^{|q_i|} r\text{-align}(q_{i,t} \to e_i) + \frac{1}{\hat{n}_i}\sum_{j=1}^{\hat{n}_i} r\text{-align}(e_i^j \to q_i)\right).$$

Informativeness (Chain) $(e_i \implies q_i)$. Measures agreement between the reasoning chain and the problem statement as a whole:

$$\text{Info-Chain}(e_i) = \frac{1 + \cos(e_i, q_i)}{2}.$$

Repetition (Step) $(\hat{s}_i^j \leftrightarrow \hat{s}_i^k)$. To identify repeated or paraphrased reasoning steps, we measure similarity between embeddings of different steps in the reasoning chain. Each step $\hat{s}_i^j$ is represented as a single embedding, and repetition is computed via cosine similarity between step embeddings:

$$\text{Repetition-Step}(e_i) = \frac{1}{2}\left[1 - \max_{j=2..\hat{n}_i} \max_{k=1..j-1} \cos(\hat{s}_i^j, \hat{s}_i^k)\right].$$

Discourse Representation $(e_i \Leftrightarrow q_i)$. Assesses whether any generated step contradicts the problem statement:

$$\text{Discourse}(e_i) = 1 - \max_{j=1..\hat{n}_i,\, t=1..|q_i|} p_{\text{contr}}(\hat{s}_i^j, q_{i,t}),$$

where p_{contr} is the contradiction probability predicted by a natural language inference (NLI) model.

Coherence (Step vs. Step). Checks for contradictions between generated steps:

$$\text{Coherence}(e_i) = 1 - \max_{j=2..\hat{n}_i,\, k<j} p_{\text{contr}}(\hat{s}_i^j, \hat{s}_i^k).$$

G Detailed Results

Table 11: **Main Results on CUMath.** Evaluation of LLMs across four prompting strategies and four metrics: Accuracy (Acc), Semantic F1, SRS, and VR. The highest value in each column is **bold and underlined**.

Model	Zero-shot				CoT				ToT				SC			
	Acc	F1	SRS	VR	Acc	F1	SRS	VR	Acc	F1	SRS	VR	Acc	F1	SRS	VR
Closed-source Models																
GPT-5.2	**0.28**	**0.44**	**0.45**	**0.01**	**0.31**	**0.42**	0.44	**-0.06**	**0.29**	**0.29**	0.36	**-0.23**	**0.29**	**0.17**	0.33	**-0.20**
GPT-4.1	0.20	0.11	0.40	-0.50	0.19	0.10	0.42	-0.45	0.19	0.03	0.42	-0.67	0.18	0.02	0.42	-0.63
OpenAI o3	0.18	0.20	**0.45**	-0.21	0.16	0.16	**0.45**	-0.22	0.17	0.04	0.42	-0.46	0.17	0.05	0.43	-0.40
Claude Sonnet 3.7	0.20	0.16	0.44	-0.51	0.19	0.15	0.44	-0.52	0.19	0.07	**0.43**	-0.64	0.17	0.10	**0.44**	-0.60
Open-source Models																
DeepSeek-R1-Distill-Qwen-32B	0.18	0.09	0.43	-0.59	0.17	0.07	0.42	-0.58	0.18	0.03	**0.43**	-0.76	0.16	0.01	0.43	-0.77
Gemma 2 9B IT	0.16	0.13	0.43	-0.37	0.16	0.18	0.44	-0.33	0.14	0.06	0.41	-0.61	0.08	0.04	0.42	-0.59
LLaMA 3 8B Instruct	0.18	0.13	0.42	-0.51	0.17	0.18	0.42	-0.51	0.18	0.05	0.42	-0.71	0.15	0.04	0.43	-0.74
LLaMA 3 70B Instruct	0.20	0.26	**0.45**	-0.28	0.18	0.25	0.44	-0.40	0.19	0.07	0.41	-0.67	0.16	0.07	0.42	-0.70
LLaMA 4 Scout 17B Instruct	0.22	0.15	0.42	-0.58	0.21	0.17	0.42	-0.55	0.21	0.08	0.41	-0.71	0.20	0.04	0.42	-0.72
Qwen2.5 7B Instruct	0.18	0.12	0.41	-0.56	0.18	0.13	0.42	-0.52	0.16	0.03	0.40	-0.69	0.16	0.03	0.42	-0.71
Mistral 7B Instruct v0.3	0.12	0.16	**0.45**	-0.17	0.12	0.20	0.43	-0.35	0.12	0.05	0.40	-0.71	0.10	0.04	0.41	-0.73
Math-specialized Models																
Qwen2.5-Math-7B Instruct	0.10	0.12	0.40	-0.58	0.08	0.07	0.40	-0.48	0.09	0.03	0.38	-0.70	0.09	0.02	0.38	-0.74
Llemma-7B	0.03	0.02	0.36	-0.39	0.03	0.02	0.36	-0.36	0.02	0.01	0.37	-0.69	0.02	0.01	0.35	-0.80
Qwen2.5-Math-1.5B Instruct	0.06	0.10	0.42	-0.38	0.07	0.06	0.39	-0.39	0.08	0.02	0.41	-0.46	0.07	0.01	0.41	-0.50
LLaMA-3.2-1B Instruct (ft)	0.07	0.15	0.41	-0.47	0.05	0.08	0.40	-0.37	0.06	0.03	0.41	-0.45	0.05	0.02	0.40	-0.51

Table 12: **Main Results on Calculus.** Evaluation of LLMs across four prompting strategies and four metrics: Accuracy (Acc), Semantic F1, SRS, and VR. The highest value in each column is **bold and underlined**.

Model	Zero-shot				CoT				ToT				SC			
	Acc	F1	SRS	VR	Acc	F1	SRS	VR	Acc	F1	SRS	VR	Acc	F1	SRS	VR
Closed-source Models																
GPT-5.2	**0.30**	**0.57**	**0.45**	**0.08**	**0.40**	**0.47**	**0.45**	**0.05**	**0.33**	**0.33**	0.37	**-0.11**	**0.31**	**0.19**	0.34	**-0.13**
GPT-4.1	0.24	0.22	0.35	-0.93	0.12	0.09	0.38	-0.61	0.24	0.05	0.35	-0.92	0.12	0.02	0.37	-0.77
OpenAI o3	0.25	0.24	0.39	-0.64	0.11	0.14	**0.45**	-0.33	0.21	0.07	0.36	-0.92	0.11	0.05	**0.41**	-0.53
Claude Sonnet 3.7	0.26	0.29	0.40	-0.53	0.11	0.19	0.41	-0.71	0.24	0.15	**0.40**	-0.83	0.11	0.16	**0.41**	-0.75
Open-source Models																
DeepSeek-R1-Distill-Qwen-32B	0.29	0.17	0.38	-0.95	0.12	0.08	0.40	-0.72	0.32	0.06	0.38	-0.96	0.12	0.02	0.37	-0.92
Gemma 2 9B IT	0.18	0.10	0.39	-0.65	0.09	0.12	0.42	-0.49	0.21	0.13	0.37	-0.94	0.03	0.04	0.39	-0.74
LLaMA 3 8B Instruct	0.21	0.08	0.41	-0.77	0.08	0.15	0.41	-0.68	0.21	0.08	0.39	-0.94	0.06	0.04	0.39	-0.91
LLaMA 3 70B Instruct	0.26	0.40	0.39	-0.52	0.09	0.22	0.41	-0.66	0.29	0.12	0.37	-0.70	0.06	0.08	0.38	-0.90
LLaMA 4 Scout 17B Instruct	0.26	0.18	0.40	-0.77	0.13	0.18	0.39	-0.75	0.26	0.20	0.38	-0.98	0.13	0.05	0.38	-0.91
Qwen2.5 7B Instruct	0.26	0.14	0.35	-0.89	0.12	0.12	0.39	-0.69	0.11	0.03	0.36	-0.81	0.11	0.03	0.37	-0.87
Mistral 7B Instruct v0.3	0.09	0.29	0.38	-0.15	0.04	0.15	0.43	-0.62	0.21	0.06	0.38	-0.92	0.01	0.05	0.39	-0.89
Math-specialized Models																
Qwen2.5-Math-7B Instruct	0.15	0.24	0.35	-0.93	0.05	0.13	0.38	-0.71	0.07	0.03	0.35	-0.86	0.06	0.02	0.34	-0.90
Llemma-7B	0.00	0.03	0.34	-0.53	0.01	0.02	0.39	-0.44	0.01	0.01	0.36	-0.83	0.01	0.01	0.33	-0.95
Qwen2.5-Math-1.5B Instruct	0.09	0.36	0.36	-0.92	0.06	0.10	0.38	-0.66	0.07	0.02	0.38	-0.40	0.03	0.01	0.38	-0.38
LLaMA-3.2-1B Instruct (ft)	0.17	0.38	0.38	-0.67	0.03	0.11	0.42	-0.62	0.01	0.02	0.38	-0.30	0.01	0.01	0.38	-0.40

Table 13: **Main Results on Differential Equations.** Evaluation of LLMs across four prompting strategies and four metrics: Accuracy (Acc), Semantic F1, SRS, and VR. The highest value in each column is **bold and underlined**.

Model	**Zero-shot**				**CoT**				**ToT**				**SC**			
	Acc	F1	SRS	VR	Acc	F1	SRS	VR	Acc	F1	SRS	VR	Acc	F1	SRS	VR
Closed-source Models																
GPT-5.2	0.08	**0.45**	0.36	-0.29	**0.11**	**0.45**	0.34	-0.36	**0.10**	**0.33**	0.30	-0.49	**0.11**	**0.20**	0.29	-0.47
GPT-4.1	0.08	0.08	0.37	-0.55	0.09	0.09	0.37	-0.53	0.07	0.03	0.36	-0.71	0.10	0.02	0.36	-0.74
OpenAI o3	**0.09**	0.16	**0.46**	**-0.20**	0.10	0.13	**0.46**	**-0.24**	0.08	0.05	**0.45**	**-0.27**	0.09	0.06	**0.41**	**-0.38**
Claude Sonnet 3.7	0.07	0.18	0.40	-0.71	0.08	0.17	0.40	-0.64	0.07	0.12	0.39	-0.74	0.07	0.17	0.40	-0.68
Open-source Models																
DeepSeek-R1-Distill-Qwen-32B	0.05	0.09	0.39	-0.65	0.06	0.09	0.38	-0.67	0.04	0.04	0.37	-0.86	0.05	0.02	0.37	-0.90
Gemma 2 9B IT	0.04	0.13	0.41	-0.45	0.05	0.18	0.42	-0.44	0.03	0.07	0.38	-0.67	0.04	0.05	0.38	-0.73
LLaMA 3 8B Instruct	0.04	0.14	0.40	-0.62	0.05	0.20	0.40	-0.60	0.04	0.05	0.38	-0.85	0.05	0.04	0.38	-0.90
LLaMA 3 70B Instruct	0.08	0.18	0.45	-0.37	0.09	0.30	0.42	-0.47	0.07	0.09	0.38	-0.81	0.08	0.09	0.38	-0.84
LLaMA 4 Scout 17B Instruct	0.05	0.17	0.39	-0.71	0.06	0.22	0.39	-0.63	0.04	0.09	0.37	-0.85	0.05	0.05	0.37	-0.90
Qwen2.5 7B Instruct	0.04	0.12	0.38	-0.62	0.05	0.14	0.39	-0.57	0.03	0.04	0.36	-0.79	0.04	0.03	0.37	-0.82
Mistral 7B Instruct v0.3	0.03	0.10	0.43	-0.47	0.04	0.20	0.42	-0.48	0.03	0.05	0.38	-0.82	0.04	0.04	0.39	-0.87
Math-specialized Models																
Qwen2.5-Math-7B Instruct	0.02	0.10	0.38	-0.62	0.03	0.05	0.39	-0.49	0.02	0.03	0.34	-0.79	0.03	0.02	0.34	-0.83
Llemma-7B	0.01	0.03	0.40	-0.38	0.02	0.03	0.38	-0.41	0.01	0.01	0.35	-0.80	0.01	0.01	0.33	-0.94
Qwen2.5-Math-1.5B Instruct	0.02	0.08	0.39	-0.56	0.02	0.09	0.38	-0.58	0.01	0.04	0.37	-0.74	0.01	0.03	0.37	-0.79
LLaMA-3.2-1B Instruct (ft)	0.02	0.05	0.41	-0.50	0.02	0.06	0.41	-0.51	0.01	0.03	0.39	-0.76	0.01	0.02	0.38	-0.83

Table 14: **Main Results on Discrete Mathematics.** Evaluation of LLMs across four prompting strategies and four metrics: Accuracy (Acc), Semantic F1, SRS, and VR. The highest value in each column is **bold and underlined**.

Model	**Zero-shot**				**CoT**				**ToT**				**SC**			
	Acc	F1	SRS	VR	Acc	F1	SRS	VR	Acc	F1	SRS	VR	Acc	F1	SRS	VR
Closed-source Models																
GPT-5.2	**0.34**	**0.31**	0.46	**-0.01**	**0.35**	**0.36**	0.47	**-0.10**	**0.36**	**0.24**	0.38	**-0.30**	**0.34**	**0.14**	0.34	**-0.25**
GPT-4.1	0.17	0.08	0.40	-0.66	0.15	0.09	0.41	-0.63	0.16	0.02	0.37	-0.87	0.16	0.02	0.37	-0.85
OpenAI o3	0.11	0.13	0.52	-0.38	0.11	0.10	0.48	-0.47	0.11	0.03	**0.41**	-0.71	0.12	0.04	0.41	-0.71
Claude Sonnet 3.7	0.16	0.10	0.43	-0.73	0.24	0.10	0.43	-0.70	0.16	0.05	**0.41**	-0.78	0.15	0.07	**0.43**	-0.75
Open-source Models																
DeepSeek-R1-Distill-Qwen-32B	0.15	0.06	0.40	-0.87	0.16	0.06	0.39	-0.88	0.15	0.02	0.38	-0.96	0.16	0.01	0.38	-0.96
Gemma 2 9B IT	0.15	0.10	0.46	-0.58	0.16	0.13	**0.50**	-0.48	0.11	0.04	**0.41**	-0.76	0.10	0.03	0.41	-0.77
LLaMA 3 8B Instruct	0.14	0.09	0.44	-0.71	0.16	0.11	0.43	-0.73	0.23	0.03	0.40	-0.87	0.11	0.03	0.40	-0.91
LLaMA 3 70B Instruct	0.18	0.14	**0.53**	-0.41	0.18	0.14	0.46	-0.58	0.19	0.04	0.40	-0.84	0.16	0.04	0.40	-0.86
LLaMA 4 Scout 17B Instruct	0.21	0.10	0.42	-0.82	0.21	0.11	0.43	-0.79	0.19	0.04	0.40	-0.88	0.20	0.03	0.40	-0.91
Qwen2.5 7B Instruct	0.17	0.09	0.40	-0.76	0.18	0.11	0.40	-0.74	0.17	0.03	0.37	-0.89	0.17	0.02	0.38	-0.91
Mistral 7B Instruct v0.3	0.08	0.09	0.51	-0.42	0.05	0.14	0.43	-0.44	0.10	0.04	**0.41**	-0.86	0.05	0.03	0.40	-0.87
Math-specialized Models																
Qwen2.5-Math-7B Instruct	0.07	0.08	0.39	-0.79	0.05	0.02	0.40	-0.61	0.06	0.02	0.35	-0.90	0.06	0.01	0.35	-0.93
Llemma-7B	0.00	0.01	0.43	-0.49	0.02	0.02	0.42	-0.45	0.01	0.00	0.38	-0.86	0.00	0.00	0.34	-0.96
Qwen2.5-Math-1.5B Instruct	0.01	0.04	0.45	-0.40	0.05	0.06	0.39	-0.77	0.07	0.02	0.38	-0.86	0.06	0.01	0.37	-0.91
LLaMA-3.2-1B Instruct (ft)	0.02	0.06	0.44	-0.68	0.01	0.07	0.44	-0.67	0.05	0.03	0.40	-0.84	0.04	0.02	0.40	-0.88

Table 15: **Main Results on Multivariable Calculus.** Evaluation of LLMs across four prompting strategies and four metrics: Accuracy (Acc), Semantic F1, SRS, and VR. The highest value in each column is **bold and underlined**.

Model	Zero-shot				CoT				ToT				SC			
	Acc	F1	SRS	VR	Acc	F1	SRS	VR	Acc	F1	SRS	VR	Acc	F1	SRS	VR
Closed-source Models																
GPT-5.2	**0.35**	**0.36**	0.41	0.05	**0.35**	**0.36**	0.43	0.03	**0.34**	**0.25**	0.37	-0.12	**0.34**	**0.13**	0.33	-0.12
GPT-4.1	0.07	0.06	0.53	0.10	0.07	0.06	**0.57**	0.10	0.07	0.01	**0.76**	0.02	0.08	0.01	0.73	0.02
OpenAI o3	0.08	0.24	0.23	0.10	0.07	0.23	0.30	0.10	0.08	0.01	0.46	0.03	0.08	0.02	0.50	0.03
Claude Sonnet 3.7	0.08	0.07	0.54	0.12	0.05	0.06	0.55	0.10	0.09	0.03	0.61	**0.06**	0.08	0.05	0.59	**0.09**
Open-source Models																
DeepSeek-R1-Distill-Qwen-32B	0.08	0.03	**0.58**	0.06	0.07	0.03	0.54	0.06	0.07	0.01	0.70	0.02	0.07	0.01	**0.75**	0.01
Gemma 2 9B IT	0.04	0.07	0.34	0.10	0.03	0.06	0.28	0.09	0.04	0.02	0.51	0.04	0.01	0.01	0.57	0.03
LLaMA 3 8B Instruct	0.05	0.07	0.41	0.11	0.03	0.05	0.39	0.09	0.02	0.01	0.57	0.03	0.04	0.01	0.65	0.02
LLaMA 3 70B Instruct	0.09	0.09	0.23	0.12	0.04	0.07	0.38	**0.12**	0.04	0.02	0.56	0.04	0.03	0.02	0.64	0.04
LLaMA 4 Scout 17B Instruct	0.13	0.06	0.49	0.11	0.11	0.06	0.49	0.11	0.10	0.02	0.57	0.04	0.13	0.01	0.64	0.03
Qwen2.5 7B Instruct	0.10	0.07	0.51	0.11	0.08	0.04	0.55	0.07	0.09	0.01	0.61	0.02	0.07	0.01	0.68	0.02
Mistral 7B Instruct v0.3	0.03	0.12	0.28	**0.15**	0.01	0.08	0.32	**0.12**	0.00	0.02	0.46	0.03	0.01	0.01	0.50	0.03
Math-specialized Models																
Qwen2.5-Math-7B Instruct	0.01	0.03	0.47	0.06	0.01	0.03	0.41	0.05	0.02	0.01	0.57	0.02	0.01	0.01	0.60	0.01
Llemma-7B	0.01	0.00	0.15	0.01	0.01	0.00	0.15	0.01	0.00	0.00	0.35	0.00	0.00	0.00	0.45	0.00
Qwen2.5-Math-1.5B Instruct	0.04	0.02	0.42	0.04	0.03	0.02	0.39	0.04	0.05	0.01	0.55	0.02	0.04	0.01	0.59	0.01
LLaMA-3.2-1B Instruct (ft)	0.01	0.04	0.29	0.06	0.01	0.04	0.27	0.06	0.01	0.01	0.47	0.02	0.01	0.01	0.47	0.01

Table 16: **Main Results on Linear Algebra.** Evaluation of LLMs across four prompting strategies and four metrics: Accuracy (Acc), Semantic F1, SRS, and VR. The highest value in each column is **bold and underlined**.

Model	Zero-shot				CoT				ToT				SC			
	Acc	F1	SRS	VR	Acc	F1	SRS	VR	Acc	F1	SRS	VR	Acc	F1	SRS	VR
Closed-source Models																
GPT-5.2	0.16	**0.31**	0.43	**0.05**	0.17	**0.29**	0.39	**-0.00**	0.17	**0.20**	0.35	**-0.10**	0.14	**0.13**	0.33	**-0.10**
GPT-4.1	0.09	0.06	0.38	-0.43	0.12	0.06	0.38	-0.44	0.08	0.02	0.37	-0.64	0.10	0.01	0.37	-0.63
OpenAI o3	0.08	0.07	**0.47**	-0.06	0.08	0.05	**0.45**	-0.08	0.07	0.02	**0.43**	-0.24	0.08	0.03	**0.41**	-0.23
Claude Sonnet 3.7	0.09	0.11	0.41	-0.52	0.09	0.11	0.41	-0.52	0.08	0.06	0.40	-0.54	0.09	0.09	**0.41**	-0.50
Open-source Models																
DeepSeek-R1-Distill-Qwen-32B	0.05	0.04	0.39	-0.57	0.11	0.04	0.39	-0.57	0.06	0.02	0.38	-0.77	0.10	0.01	0.37	-0.82
Gemma 2 9B IT	0.10	0.10	0.42	-0.34	0.14	0.18	0.44	-0.30	0.08	0.04	0.40	-0.55	0.05	0.04	0.39	-0.62
LLaMA 3 8B Instruct	0.19	0.13	0.42	-0.46	0.20	0.19	0.41	-0.50	0.19	0.04	0.39	-0.68	**0.21**	0.03	0.38	-0.76
LLaMA 3 70B Instruct	0.12	0.17	0.44	-0.37	0.15	0.26	0.44	-0.43	0.11	0.05	0.39	-0.75	0.10	0.05	0.39	-0.72
LLaMA 4 Scout 17B Instruct	**0.21**	0.10	0.40	-0.57	**0.22**	0.14	0.40	-0.53	**0.21**	0.04	0.38	-0.71	**0.21**	0.03	0.38	-0.74
Qwen2.5 7B Instruct	0.09	0.10	0.39	-0.56	0.12	0.10	0.39	-0.56	0.09	0.03	0.37	-0.71	0.10	0.03	0.37	-0.75
Mistral 7B Instruct v0.3	0.10	0.12	0.43	-0.38	0.15	0.22	**0.45**	-0.34	0.06	0.05	0.39	-0.73	0.08	0.04	0.39	-0.78
Math-specialized Models																
Qwen2.5-Math-7B Instruct	0.02	0.07	0.38	-0.62	0.04	0.04	0.41	-0.46	0.03	0.02	0.34	-0.80	0.02	0.01	0.34	-0.84
Llemma-7B	0.01	0.03	0.39	-0.34	0.03	0.03	0.39	-0.35	0.01	0.01	0.36	-0.69	0.01	0.01	0.34	-0.85
Qwen2.5-Math-1.5B Instruct	0.01	0.02	0.46	-0.04	0.01	0.04	0.41	-0.36	0.02	0.02	0.37	-0.71	0.02	0.01	0.36	-0.76
LLaMA-3.2-1B Instruct (ft)	0.01	0.10	0.42	-0.41	0.05	0.08	0.40	-0.49	0.05	0.04	0.39	-0.66	0.04	0.03	0.38	-0.75

Table 17: **Main Results on Pre-calculus.** Evaluation of LLMs across four prompting strategies and four metrics: Accuracy (Acc), Semantic F1, SRS, and VR. The highest value in each column is **bold and underlined**.

Model	Zero-shot				CoT				ToT				SC			
	Acc	F1	SRS	VR	Acc	F1	SRS	VR	Acc	F1	SRS	VR	Acc	F1	SRS	VR
Closed-source Models																
GPT-5.2	**0.47**	**0.51**	0.53	**0.09**	**0.51**	**0.51**	**0.53**	**-0.03**	**0.48**	**0.29**	0.37	**-0.25**	**0.49**	**0.20**	0.35	**-0.15**
GPT-4.1	0.38	0.13	0.41	-0.51	0.40	0.15	0.41	-0.53	0.36	0.03	0.37	-0.79	0.38	0.03	0.38	-0.70
OpenAI o3	0.32	0.29	**0.55**	-0.25	0.33	0.24	0.51	-0.36	0.30	0.05	**0.42**	-0.61	0.34	0.07	**0.43**	-0.55
Claude Sonnet 3.7	0.38	0.21	0.43	-0.73	0.41	0.23	0.44	-0.66	0.36	0.06	0.41	-0.83	0.38	0.12	0.42	-0.79
Open-source Models																
DeepSeek-R1-Distill-Qwen-32B	0.39	0.10	0.41	-0.71	0.44	0.10	0.40	-0.70	0.39	0.03	0.38	-0.90	0.36	0.02	0.38	-0.87
Gemma 2 9B IT	0.34	0.19	0.48	-0.43	0.37	0.36	0.52	-0.42	0.23	0.07	0.40	-0.77	0.08	0.05	0.42	-0.69
LLaMA 3 8B Instruct	0.39	0.22	0.44	-0.64	0.40	0.27	0.45	-0.64	0.36	0.06	0.40	-0.86	0.36	0.05	0.40	-0.87
LLaMA 3 70B Instruct	0.38	0.33	0.54	-0.32	0.41	0.40	0.51	-0.44	0.38	0.08	0.39	-0.83	0.39	0.08	0.40	-0.82
LLaMA 4 Scout 17B Instruct	0.42	0.20	0.43	-0.70	0.44	0.23	0.43	-0.69	0.43	0.08	0.40	-0.82	0.40	0.05	0.40	-0.80
Qwen2.5 7B Instruct	0.39	0.15	0.42	-0.63	0.44	0.21	0.42	-0.63	0.40	0.04	0.37	-0.83	0.40	0.04	0.38	-0.84
Mistral 7B Instruct v0.3	0.20	0.19	0.54	-0.16	0.30	0.33	0.50	-0.39	0.18	0.06	0.41	-0.81	0.24	0.07	0.40	-0.86
Math-specialized Models																
Qwen2.5-Math-7B Instruct	0.25	0.13	0.40	-0.64	0.24	0.10	0.40	-0.64	0.25	0.04	0.38	-0.77	0.27	0.03	0.37	-0.83
Llemma-7B	0.05	0.02	0.41	-0.58	0.04	0.02	0.41	-0.56	0.03	0.01	0.38	-0.84	0.03	0.01	0.34	-0.95
Qwen2.5-Math-1.5B Instruct	0.10	0.07	0.46	-0.39	0.22	0.10	0.41	-0.60	0.22	0.04	0.39	-0.76	0.22	0.03	0.38	-0.81
LLaMA-3.2-1B Instruct (ft)	0.12	0.20	0.45	-0.59	0.11	0.21	0.45	-0.62	0.14	0.06	0.41	-0.83	0.12	0.05	0.41	-0.85

Table 18: **Main Results on Trigonometry.** Evaluation of LLMs across four prompting strategies and four metrics: Accuracy (Acc), Semantic F1, SRS, and VR. The highest value in each column is **bold and underlined**.

Model	Zero-shot				CoT				ToT				SC			
	Acc	F1	SRS	VR	Acc	F1	SRS	VR	Acc	F1	SRS	VR	Acc	F1	SRS	VR
Closed-source Models																
GPT-5.2	0.28	**0.57**	0.51	0.12	0.29	**0.52**	**0.49**	-0.02	0.28	**0.35**	0.37	-0.23	0.28	**0.22**	0.34	-0.21
GPT-4.1	0.35	0.12	0.39	-0.53	**0.35**	0.13	0.40	-0.53	**0.34**	0.02	0.37	-0.76	**0.34**	0.02	0.37	-0.73
OpenAI o3	**0.36**	0.24	0.52	-0.02	**0.35**	0.20	0.48	-0.17	**0.34**	0.04	**0.43**	-0.48	**0.34**	0.06	**0.43**	-0.41
Claude Sonnet 3.7	0.33	0.18	0.44	-0.50	0.33	0.18	0.44	-0.49	0.31	0.05	0.40	-0.80	0.31	0.06	0.40	-0.79
Open-source Models																
DeepSeek-R1-Distill-Qwen-32B	0.26	0.14	0.44	-0.45	0.25	0.10	0.42	-0.61	0.24	0.03	0.40	-0.89	0.24	0.01	0.39	-0.92
Gemma 2 9B IT	0.27	0.19	0.48	-0.24	0.28	0.26	0.47	-0.30	0.25	0.06	0.41	-0.65	0.24	0.04	0.41	-0.62
LLaMA 3 8B Instruct	0.26	0.21	0.43	-0.45	0.25	0.26	0.43	-0.49	0.24	0.05	0.40	-0.83	0.25	0.05	0.40	-0.83
LLaMA 3 70B Instruct	0.29	0.50	0.54	-0.11	0.28	0.38	0.47	-0.33	0.27	0.09	0.38	-0.81	0.28	0.10	0.38	-0.80
LLaMA 4 Scout 17B Instruct	0.27	0.21	0.42	-0.59	0.27	0.24	0.42	-0.58	0.26	0.08	0.39	-0.79	0.26	0.05	0.39	-0.84
Qwen2.5 7B Instruct	0.24	0.19	0.41	-0.57	0.24	0.18	0.42	-0.54	0.23	0.03	0.38	-0.80	0.23	0.04	0.38	-0.82
Mistral 7B Instruct v0.3	0.28	0.19	**0.57**	**0.21**	0.27	0.30	0.47	-0.32	0.25	0.06	0.40	-0.86	0.25	0.06	0.40	-0.86
Math-specialized Models																
Qwen2.5-Math-7B Instruct	0.18	0.18	0.41	-0.55	0.17	0.09	0.41	-0.52	0.16	0.05	0.36	-0.80	0.15	0.02	0.35	-0.86
Llemma-7B	0.12	0.03	0.40	-0.41	0.11	0.02	0.40	-0.29	0.10	0.01	0.38	-0.81	0.10	0.01	0.35	-0.93
Qwen2.5-Math-1.5B Instruct	0.12	0.11	0.43	-0.38	0.10	0.00	0.40	**0.23**	0.10	0.00	0.40	**0.23**	0.10	0.00	0.40	**0.15**
LLaMA-3.2-1B Instruct (ft)	0.14	0.20	0.45	-0.51	0.13	0.00	0.40	**0.23**	0.12	0.00	0.40	**0.23**	0.11	0.00	0.40	**0.15**

H LLM Solutions

In the following solution transcripts, any text highlighted in red indicates a step or claim where the model's reasoning is incorrect.

Solution 1: GPT-4.1 Solution (Indefinite Integral)

We are given the integral:

$$\int \frac{1-\sin x}{x+\cos x}\,dx$$

Step 1: Use substitution Let:

$$u = x + \cos x$$

Then:

$$\frac{du}{dx} = 1 - \sin x \quad \Rightarrow \quad du = (1-\sin x)\,dx$$

This matches the numerator exactly, so we can rewrite the integral as:

$$\int \frac{1-\sin x}{x+\cos x}\,dx = \int \frac{1}{u}\,du$$

Step 2: Integrate

$$\int \frac{1}{u}\,du = \ln|u| + C = \ln|x+\cos x| + C$$

Final Answer

$$\boxed{\int \frac{1-\sin x}{x+\cos x}\,dx = \ln|x+\cos x| + C}$$

Solution 2: GPT-4.1 Solution (Definite Integral)

We are given the definite integral:

$$I = \int_{-\pi/6}^{\pi/6} \frac{1-\sin x}{x+\cos x}\,dx.$$

—

Step 1: Consider symmetry
The interval is symmetric about 0: from $-\frac{\pi}{6}$ to $\frac{\pi}{6}$. Let's define the integrand:

$$f(x) = \frac{1-\sin x}{x+\cos x}.$$

To evaluate this, let's analyze its **symmetry** by examining $f(-x)$:

$$f(-x) = \frac{1-\sin(-x)}{-x+\cos(-x)} = \frac{1+\sin x}{-x+\cos x}.$$

So,

$$f(-x) = \frac{1+\sin x}{-x+\cos x}.$$

This doesn't look like an even or odd function directly. But we **add** $f(x)$ and $f(-x)$ to explore further:

—

Step 2: Add $f(x) + f(-x)$

$$f(x)+f(-x) = \frac{1-\sin x}{x+\cos x} + \frac{1+\sin x}{-x+\cos x}.$$

Let's write it as a single expression:
Let's denote:

$$A = \frac{1-\sin x}{x+\cos x}, \quad B = \frac{1+\sin x}{-x+\cos x}.$$

Common denominator is:

$$(x+\cos x)(-x+\cos x) = \cos^2 x - x^2.$$

So we get:

$$A+B$$

$$= \frac{(1-\sin x)(-x+\cos x) + (1+\sin x)(x+\cos x)}{\cos^2 x - x^2}.$$

Now expand numerator:

- $(1-\sin x)(-x+\cos x) = -x(1-\sin x) + \cos x(1-\sin x) = -x + x\sin x + \cos x - \cos x \sin x$
- $(1+\sin x)(x+\cos x) = x(1+\sin x) + \cos x(1+\sin x) = x + x\sin x + \cos x + \cos x \sin x$

Add both:

$$\text{Numerator}$$

$$\begin{aligned} &= (-x + x\sin x + \cos x - \cos x\sin x) \\ &\quad + (x + x\sin x + \cos x + \cos x \sin x) \end{aligned}$$

Simplify:

- $-x + x = 0$
- $x\sin x + x\sin x = 2x\sin x$

- $\cos x + \cos x = 2\cos x$
- $-\cos x \sin x + \cos x \sin x = 0$

So total numerator is:

$$2x\sin x + 2\cos x = 2(x\sin x + \cos x).$$

Therefore,

$$f(x) + f(-x) = \frac{2(x\sin x + \cos x)}{\cos^2 x - x^2}.$$

—
Step 3: Now integrate $f(x) + f(-x)$ over $-\pi/6$ to $\pi/6$
From integral properties over symmetric intervals:

$$\int_{-a}^{a} f(x)\,dx = \int_{-a}^{a} \frac{1}{2}[f(x) + f(-x)]\,dx.$$

So:

$$\begin{aligned} I &= \int_{-\pi/6}^{\pi/6} f(x)\,dx \\ &= \frac{1}{2}\int_{-\pi/6}^{\pi/6} \big(f(x) + f(-x)\big)\,dx \\ &= \frac{1}{2}\int_{-\pi/6}^{\pi/6} \frac{2(x\sin x + \cos x)}{\cos^2 x - x^2}\,dx \end{aligned}$$

Simplify:

$$I = \int_{-\pi/6}^{\pi/6} \frac{x\sin x + \cos x}{\cos^2 x - x^2}\,dx.$$

—
Step 4: Check whether the integrand is **odd**
Let:

$$g(x) = \frac{x\sin x + \cos x}{\cos^2 x - x^2}.$$

Compute $g(-x)$:

- Numerator: $-x\sin x + \cos x$
- Denominator: $\cos^2 x - x^2$ (even function)

So,

$$\begin{aligned} g(-x) &= \frac{-x\sin x + \cos x}{\cos^2 x - x^2} \\ &= -\frac{x\sin x + \cos x}{\cos^2 x - x^2} \\ &= -g(x) \end{aligned}$$

Thus, $g(x)$ is an odd function.

—
Step 5: Integral of an odd function over symmetric interval
Since $g(x)$ is odd and the limits are symmetric:

$$\int_{-a}^{a} g(x)\,dx = 0$$

So,

$$I = \int_{-\pi/6}^{\pi/6} f(x)\,dx = \int_{-\pi/6}^{\pi/6} g(x)\,dx = 0$$

—
Final Answer:

$$\boxed{0}$$

Solution 3: GPT-5.2 Solution (Definite Integral)

$$\text{Let } I = \int_{-\pi/6}^{\pi/6} \frac{1-\sin x}{x+\cos x}\,dx.$$

Use the substitution $x \mapsto -x$:

$$\begin{aligned} I &= \int_{-\pi/6}^{\pi/6} \frac{1-\sin(-x)}{-x+\cos(-x)}\,dx \\ &= \int_{-\pi/6}^{\pi/6} \frac{1+\sin x}{\cos x - x}\,dx. \end{aligned}$$

Add the two expressions for I:

$$2I = \int_{-\pi/6}^{\pi/6} \left(\frac{1-\sin x}{x+\cos x} + \frac{1+\sin x}{\cos x - x}\right)dx.$$

With common denominator

$$(x+\cos x)(\cos x - x) = \cos^2 x - x^2,$$

the numerator is

$$\begin{aligned}(1-\sin x)&(\cos x - x)\\ &+(1+\sin x)(x+\cos x)\\ &= 2\cos x + 2x\sin x.\end{aligned}$$

Hence

$$\begin{aligned}2I &= \int_{-\pi/6}^{\pi/6} \frac{2(\cos x + x\sin x)}{\cos^2 x - x^2}\,dx\\ &= \int_{-\pi/6}^{\pi/6} \frac{\cos x + x\sin x}{\cos^2 x - x^2}\,dx.\end{aligned}$$

Note

$$\frac{d}{dx}(\cos^2 x - x^2) = -2(\cos x\sin x + x),$$

so

$$\frac{d}{dx}\ln(\cos^2 x - x^2) = \frac{-2(\cos x\sin x + x)}{\cos^2 x - x^2}.$$

Also,

$$\begin{aligned}\frac{\cos x + x\sin x}{\cos^2 x - x^2} &= -\frac{1}{2}\frac{d}{dx}\ln(\cos^2 x - x^2)\\ &\quad\cdot \frac{1}{\sin x} \qquad \text{(not convenient)},\end{aligned}$$

but observe instead that

$$\begin{aligned}&\frac{d}{dx}\ln\left(\frac{\cos x + x}{\cos x - x}\right)\\ &= \frac{-\sin x + 1}{\cos x + x} - \frac{-\sin x - 1}{\cos x - x}\\ &= \frac{1-\sin x}{x+\cos x} + \frac{1+\sin x}{\cos x - x}.\end{aligned}$$

Therefore,

$$2I = \left[\ln\left(\frac{\cos x + x}{\cos x - x}\right)\right]_{-\pi/6}^{\pi/6}.$$

Compute endpoints. Since

$$\cos(\pi/6) = \cos(-\pi/6) = \frac{\sqrt{3}}{2},$$

we get

$$\begin{aligned}2I &= \ln\left(\frac{\frac{\sqrt{3}}{2}+\frac{\pi}{6}}{\frac{\sqrt{3}}{2}-\frac{\pi}{6}}\right)\\ &\quad - \ln\left(\frac{\frac{\sqrt{3}}{2}-\frac{\pi}{6}}{\frac{\sqrt{3}}{2}+\frac{\pi}{6}}\right)\\ &= 2\ln\left(\frac{\frac{\sqrt{3}}{2}+\frac{\pi}{6}}{\frac{\sqrt{3}}{2}-\frac{\pi}{6}}\right).\end{aligned}$$

Thus

$$\begin{aligned}I &= \ln\left(\frac{\frac{\sqrt{3}}{2}+\frac{\pi}{6}}{\frac{\sqrt{3}}{2}-\frac{\pi}{6}}\right)\\ &= \ln\left(\frac{3\sqrt{3}+\pi}{3\sqrt{3}-\pi}\right).\end{aligned}$$

$$\boxed{\ln\left(\frac{3\sqrt{3}+\pi}{3\sqrt{3}-\pi}\right)}$$

Solution 4: OpenAI-o3 Solution (Definite Integral)

We are asked to evaluate the definite integral

$$\int_{-\pi/6}^{\pi/6} \frac{1-\sin(x)}{x+\cos(x)}\,dx.$$

First, observe that this integrand does not have a standard elementary antiderivative due to the complexity of the expression and the interaction between the trigonometric and algebraic terms. Therefore, symbolic integration is not practical.

Next, we examine the symmetry of the integrand. Define

$$f(x) = \frac{1-\sin(x)}{x+\cos(x)}.$$

Evaluating $f(-x)$ gives:

$$f(-x) = \frac{1+\sin(x)}{-x+\cos(x)},$$

which is not equal to $f(x)$ or $-f(x)$, so the function is neither even nor odd. Thus, no symmetry can be used to simplify the computation.

Since an analytical solution is unavailable, we use a numerical method. Specifically, we apply the Trapezoidal Rule with $n = 6$ subintervals over the interval $[-\pi/6, \pi/6]$. The width of each subinterval is

$$h = \frac{\pi/3}{6} = \frac{\pi}{18}.$$

Using the Trapezoidal Rule:

$$\begin{aligned}\int_a^b f(x)\,dx \approx \frac{h}{2}\Big[&f(x_0) + 2f(x_1) + 2f(x_2)\\ &+ \cdots + 2f(x_{n-1}) + f(x_n)\Big]\end{aligned}$$

After performing the calculation, we find the approximate value of the integral to be:

$$\int_{-\pi/6}^{\pi/6} \frac{1-\sin(x)}{x+\cos(x)}\,dx \approx 1.4511.$$

Therefore, the final answer is approximately

$\boxed{1.4511}$.

Solution 5: Mistral 7B Instruct v0.3 Solution (Definite Integral)

To solve this integral, we can't use a simple substitution or integration by parts, so we'll use numerical methods to approximate the value. We'll use the trapezoidal rule, which is a numerical integration method that provides a good approximation for definite integrals.

The trapezoidal rule formula is:

$$T = h * [f(a) + 2 * \Sigma(f(x_i) \\ \text{for i = 1 to n}) + f(b)]/n$$

where $h = (b-a)/n$, n is the number of subintervals, and $f(x)$ is the function we're integrating.

Let's choose $n = 1000$ for a good approximation.

First, let's find h:

$$h = (\pi/6 - (-\pi/6))/1000 \\ = \pi/(3000 * 2) = \pi/6000$$

Now, let's find the subintervals' endpoints:

$$x_i = -\pi/6 + i * h \quad \text{for i = 0 to 1000}$$

Next, we'll calculate the function values for each subinterval:

$$f(x_i) = (1-\sin(x_i))/(x_i + \cos(x_i)) \\ \text{for i = 0 to 1000}$$

Finally, we'll sum up the function values and add the endpoints' values:

$$T = h * [f(-\pi/6) + 2 * \Sigma(f(x_i) \\ \text{for i = 1 to 1000}) + f(\pi/6)]$$

After calculating the sum and multiplying by h, we get the approximate value of the integral:

$$\boxed{T \approx 1.400731}$$

This is the approximate value of the definite integral $\int_{-\pi/6}^{\pi/6} \frac{1-\sin x}{x+\cos x}\,dx$ using the trapezoidal rule with n = 1000.